Smart Farming Using Machine Learning, Deep Learning and IoT

PROJECT REPORT

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CERTIFICATE

This is to certify that the project design document entitled **Smart Farming Using Machine Learning, Deep Learning and IoT** is a bonafide record of the Project work presented by **Shegin Ismail A P** (IDK20IT038), **Sarath Sreedhar J** (IDK20IT037), **Vishnu C N** (IDK20IT046) & **Maheswari S M** (IDK20IT025) in partial fulfillment of the requirements for the award of the Degree of **Bachelor of Technology** in **Information Technology** of APJ Abdul Kalam Technological University, Kerala.

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We hereby declare that the project report **Smart Farming Using Machine Learning, Deep Learning and IoT**, being submitted in partial fulfillment for the award of B.Tech degree is the original work carried out by us. It has not formed the part of any other thesis submitted for award of any degree or diploma, either in this or any other University.

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ABSTRACT

Smart Farming epitomizes the integration of technology into agriculture, utilizing IoT, Machine Learning, and Deep Learning to boost productivity and sustainability. The proposed project introduces an advanced mobile application designed to empower farmers with intelligent insights for precision agriculture. With capabilities such as crop prediction, weed and pest detection, and environmental control, the system aims to optimize crop cultivation, reduce resource usage, and enable proactive management. Tailored for non-technical users, especially those in the farming community, the application provides a user-friendly interface for seamless interaction. The scope encompasses crucial agricultural aspects, including greenhouse automation, soil moisture control, and crop-specific recommendations. Despite its potential, the project acknowledges certain constraints. automated actions like herbicide spraying and harvesting are not covered under this project, considering constraints in resources, time, and workspace. However, the proposed system is not static; it looks toward the future with a vision for expansion. With a commitment to revolutionizing farming practices, the project envisions widespread application in farming communities, educational institutions, and government initiatives, contributing to efficient data management and sustainable agricultural practices. As technology continues to evolve, the proposed project aid in facilitating the seamless integration of cutting-edge technologies into the agricultural landscape for a more resilient and productive future.

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ABBREVIATIONS

- **pH**:Potential of Hydrogen
- **Wi-Fi**:Wireless Fidelity
- **IoT**:Internet of Things
- **GDD**:Growing Degree Days
- **GPRS** : General Packet Radio Service
- **HTTP** :Hypertext Transfer Protocol
- **GPS** : Global Positioning System
- **UI** : User Interface
- **API**:Application Programming Interface
- **SDK**:Software Development Kit
- **GDD**:Growing Degree Days
- **N**: Nitrogen
- **P**: Phosphorus
- **K**:Pottassium
- **ML**:Machine Learning
- **DL**: Deep Learning
- **NodeMCU**: Node Microcontroller Unit

CHAPTER 1

INTRODUCTION

In the pivotal realm of agriculture, the significance of technology-driven Smart Farming cannot be overstated. The proposed project aims transformative approach stands at the forefront of agricultural innovation, integrating advanced technologies such as the Internet of Things (IoT), Machine Learning, and Deep Learning. In an era where data-driven decision-making is paramount, Smart Farming leverages intelligent systems to revolutionize traditional farming practices, optimizing crop cultivation, reducing resource consumption, and empowering farmers with real-time insights. The project contributes to this paradigm shift through the development of a sophisticated mobile application. Addressing critical challenges like food insecurity and enhancing small-scale farm productivity, the system focuses on essential features such as crop prediction, weed and pest detection, and environmental control. By harnessing data collected from the farming environment, Smart Farming aims to enhance crop yield, reduce resource consumption, and enable proactive management of environmental conditions. Notably, the system is designed to cater to non-technical users, particularly farmers. This application serves as an intuitive interface for farmers, allowing them to seamlessly monitor and control various aspects of their agricultural operations.

CHAPTER 2

LITERATURE REVIEW

2.1 LITERATURE SURVEY

The literature survey provides a comprehensive exploration of the integration of Machine Learning, Deep learning and the Internet of Things (IoT) in precision agriculture, shedding light on the advancements and influences of various research studies. Saranya *et al.*[1] exemplifies the positive implications of utilizing deep learning and IoT for precision agriculture, offering valuable insights. The project design under consideration draws significant inspiration from Durai *et al.*[2] system, prompting an in-depth analysis of the algorithms employed across different modules. Banu[3], Devi *et al.*[4], Narbhaev *et al.*[5], and Unal *et al.*[6] contribute various algorithmic methodologies,to improve precision agriculture. Respective papers by Lanitha *et al.*[7] and Oteyo *et al.*[8] focus on the survey of IoT for Greenhouse and the integration of Mobile Applications for Smart Agriculture, contributing standardization and enhancements to the existing body of knowledge.

2.1.1 Integration of DL and IoT in Smart Agriculture

Saranya *et al.* [1] proposed system highlighting the integration of Deep Learning (DL) and the Internet of Things (IoT) in smart agriculture, emphasizing their diverse applications. Expanding on fundamental DL techniques, such as Convolutional Neural Networks (CNN) and Generative Adversarial Networks (GAN), the paper explores

their role in precise pest detection. Additionally, it delves into the architecture of IoT in agriculture, the integration of various sensors, and the use of Unmanned Aerial Vehicles (UAVs) for remote sensing. Addressing the pivotal role of data, the survey emphasizes data acquisition, annotation, and augmentation in agricultural datasets for effective DL model performance. The identified challenges in DL and IoT for agriculture underscore the need for comprehensive solutions. The literature also underscores how AI, particularly DL and IoT, contributes to sustainable agriculture by automating processes and enhancing productivity. Smart agriculture is presented as a strategic management approach, seamlessly integrating cutting-edge technologies to enable automatic tracking, monitoring, intelligent control, and decision-making in agriculture. Overall, the methodology advocates for the integration of DL and IoT as a promising solution to revolutionize agriculture, ensuring sustainability and meeting the demands of a growing global population.

2.1.2 Smart Farming using Machine Learning and Deep Learning

The proposed system by Durai *et al.*[2] presents a comprehensive and integrated approach to precision agriculture, addressing critical facets such as crop prediction, weed identification, pest recognition, and cost estimation. Leveraging machine learning and deep learning techniques, the study establishes a robust foundation for informed decision-making in farming practices. The crop prediction module utilizes ensemble algorithms like XGBoost and Random Forest, integrating weather data for precise recommendations. Deep learning, is employed for accurate weed and pest identification, laying the groundwork for effective management strategies. The paper further delves into cost estimation, employing regression models to analyze operational and fixed costs. Unique to this study is the incorporation of cost concepts, categorizing expenses and enhancing the understanding of cultivation economics.

1. **Crop Prediction**

   The described methodology outlines a systematic approach to crop recommendation, employing a diverse set of machine learning algorithms for predictive modeling. Beginning with data preprocessing and exploratory analysis, ensure the dataset's readiness for training. Visualization techniques aid in understanding the dataset and outlier detection mechanisms are employed judiciously, preserving outliers relevant to crop growth. Label encoding facilitates the transition to numerical values, paving the way for model training. A suite of ten classification algorithms, including Naïve Bayes, Logistic Regression, and ensemble methods like Random Forest and XGBoost, undergo rigorous evaluation and hyperparameter tuning, ensuring optimal model performance. The Random Forest Algorithm along hyperparameter tuned with randomized Search CV showed better performance. In practice, the model engages with user provided climate and soil data. The user receives a tailored list of the top 5 crop recommendations, along with essential details like Growing Degree Days and nutrient concentration requirements. This comprehensive module empowers farmers with actionable insights, enabling them to make informed decisions for successful and sustainable crop cultivation.

2. **Weed Identification**

   The Weed Identification module employs a multi step process to effectively categorize and recommend herbicides based on deep learning techniques. Beginning with the import of essential libraries and utilizing dataset from Kaggle, the module employs data augmentation to enhance the training set. The ResNet15V2 pre-trained model is then harnessed. For herbicide recommendation, a dataset with detailed herbicide information is compiled. Here a Random Forest Classifier is employed,utilizing parameters such as the user entered crop name and the weed name identified from uploaded images. Then classifier assigns probabilities to crops, and suitable herbicides are subsequently

recommended. This module not only achieves accurate weed identification through deep learning but also provides actionable insights by suggesting herbicides tailored to specific weed types and crops, offering valuable support for agricultural decision-making.

3. **Pest Identification**

   The Pest Identification and Pesticides Recommendation module follows a comprehensive process integrating deep learning and data preprocessing techniques. It initiates with the import of necessary libraries and the utilization of a pests dataset from Kaggle containing images of various insects in contact with crops. Through the division of the dataset into training, testing, and validation sets, coupled with data augmentation, the model is enriched for robust training. Leveraging the same DL model in weed pre-trained model, the module constructs a network, addressing gradient challenges and achieving an impressive accuracy.For Pesticide Recommendation, a dataset with detailed pesticide information is compiled, encompassing well suited crops and active ingredients. Here a Random Forest Classifier is deployed, utilizing parameters like user entered crop names and predicted insect names from images. Then assigns probabilities to crops, and suitable pesticides are recommended. This module excels in identifying and categorizing insects in contact with crops, utilizing deep learning, and goes further to provide actionable insights by suggesting pesticides tailored to specific insect types and crops, enhancing pest control and supporting effective crop protection in agriculture.

4. **Cost Estimation**

   The Cost Estimation module aims to predict crucial cost concepts and total cultivation expenses, providing valuable insights for addressing crises, natural calamities, and price fluctuations in cultivation resources. Leveraging Indian cost of cultivation survey data spanning from 2010 to 2018, the dataset is struc-

tured with item wise breakup attributes and cost of cultivation attributes. The initial step involves importing necessary libraries, including NumPy, Pandas, Train-test split, Label Encoder, and Machine Learning Ensemble Regressors. Subsequently, data preprocessing involves aggregating datasets, handling non-numerical columns through techniques like Label encoding, and building regression models for each item wise attribute using Machine Learning Ensemble Regressors. The independent variables, including year, state, and crop attributes, are fed into the models, enabling the prediction of operational and fixed costs. The selection of the best models is based on evaluation metrics like r2 score, root mean squared error, mean squared error, and mean absolute error. These models are then stored in a pickle file for future use, and the forecasted values until 2028 are applied to calculate operational cost, fixed cost, total cost, and various cost concepts (A1, A2, B1, B2, C1, C2, C3). The resulting data frame is converted into a Comma-separated values(CSV) file, providing a dynamic resource for future calculations and aiding farmers and policymakers in making informed decisions about crop planning and resource allocation.

2.1.3 Crop Prediction using RFC

Banu *et al.*[3] presents comprehensive overview of the significance of predicting crop yields in the context of Indian agriculture, emphasizing the crucial role accurate information plays in optimizing production and reducing losses. The introduction highlights the potential misuse of technology and introduces machine learning algorithms as a solution for predicting crop yields based on various environmental factors. The literature review draws on existing research, citing studies that underscore the effectiveness of machine learning approaches, including Random Forest and AdaBoost, in yield prediction. Geospatial techniques and remote sensors are also acknowledged for addressing crop growth variations.The methodology section details the process of data pre-processing, considering factors like temperature, rainfall, and humidity, and emphasizes the selection of machine learning algorithms. Random Forest emerges as the most accurate choice, achieving a reported accuracy rate of

91.34%. The proposed system introduces a mobile application leveraging Random Forest predictions to assist farmers in optimizing crop yields and profits.

The conclusion underscores the importance of accurate crop yield predictions for diverse Indian districts, emphasizing the potential for improved models to enhance production and land utilization.

2.1.4 Weed and Pest Identification using MobileNetV2

The paper by Nguyen *[*4]introduces MobilenetV2, a fresh strategy for object detection, emphasizing its applicability to outdoor images while preserving both speed and precision. Utilizing the MobileNetV2 structure, known for its efficiency, the research begins by generating initial feature maps. This module constructs feature maps at multiple scales from a single input image resolution, facilitating the detection of objects across various size ranges, which is crucial for real-time applications.At the heart of the proposed framework lies the detection network, consisting of a classification subnet and a box regression subnet, intricately connected to each level of the feature pyramid. The classification subnet functions by forecasting the likelihood of object presence at every location within the feature pyramid, aided by an anchor box and object class. Meanwhile, the box regression subnet focuses on determining the deviation between each anchor box and its nearby ground-truth object. This dual-subnet setup, combined with the multi-scale capabilities of the feature pyramid, ensures robust and precise object detection across diverse environmental settings and object dimensions.A key factor contributing to the success of the framework is the adoption of the focal loss function, strategically replacing the traditional cross-entropy loss. By addressing the issue of class imbalance, especially prevalent in complex outdoor backgrounds, the focal loss function enhances the training of the classification subnet. This improvement enables better differentiation between easy and hard training samples, leading to enhanced detection performance.

2.1.5 Cost Estimation using XGBoost Regressor

The paper by Narbaev *et al.*[5] discusses the utilization of the XGBoost Regressor in project management, particularly for cost forecasting, presenting it as a superior alternative to traditional Earned Value Management (EVM) models and other machine learning approaches. Emphasizing the importance of accurate cost predictions in project management, the paper introduces XGBoost as a potent tool, critiquing conventional methods for their limitations. The study's contributions include establishing an ML framework for cost forecasting, validating the reliability of the XGBoost model, and employing real project data for testing and improvement. Methodologically, the research integrates machine learning, specifically the XGBoost algorithm, into traditional EVM cost forecasting, leveraging non-linear regression features for more effective cost data handling. The findings showcase the consistent out performance of XGBoost over EVM and other machine learning models across different project stages. While the primary focus is on project management, the principles of accurate cost estimation and the advantages of machine learning can potentially extend to applications in crop cost estimation. The paper concludes by highlighting the critical nature of cost forecasting in project management and suggesting future directions for research, encouraging exploration of diverse project datasets and integration of cluster analysis for different project types.

2.1.6 Deep Learning for Object Detection

The paper by Unal *et al.*[6] explores the transformative role of deep learning (DL), internet of things (IoT), and machine learning in agriculture, with a specific focus on smart agriculture. It emphasizes the significance of remote and precise monitoring for enhancing cost-efficiency and sustainability in agricultural practices. DL, as a subset of machine learning, is highlighted for its ability to process complex data like images in applications such as disease detection, plant classification, and pest recognition. Specifically, in the context of pest recognition, the text details studies that employ deep learning techniques, such as deep neural networks and convolutional

neural networks, for the identification and classification of agricultural pests. Notable applications include pest identification in complex backgrounds, classification of moth images, detection and identification of stored-grain insects, and the development of automated systems for monitoring pests. The challenges and opportunities associated with using unmanned aerial vehicles (UAVs) in agriculture are also discussed. The study concludes by recognizing the exponential growth in DL literature related to agriculture and encourages further exploration of DL techniques, particularly in the context of pest detection and smart farming applications.

2.1.7 IoT enabled Greenhouse

The paper by B.Lanitha *et al.*[7] emphasizes the pivotal role of IoT (Internet of Things) and automation in greenhouse agriculture, providing a comprehensive overview of their applications and benefits. It emphasizes the transition from conventional farming to cloud-based and wireless architectures, enabling real-time monitoring and control of greenhouse environments. Automation, facilitated by IoT, enhances energy efficiency by ensuring optimal operation of sensors and devices, thereby addressing a significant challenge in agriculture. The integration of AI algorithms for real-time control based on comprehensive data, including climate and solar radiation, is highlighted in the paper. The methodology emphasizes the role of these technologies in disease prevention, with IoT-based sensor data predicting and preventing crop diseases. Furthermore, the paper discusses the application of AI in various agricultural domains, such as urban farming, livestock management, disease detection, and weather forecasting. Sustainable manufacturing in agriculture is explored through the application of machine learning and AI. The significance of smart energy management systems and environmental monitoring using IoT-based data collection is underlined. The paper concludes by presenting results that demonstrate the effectiveness of the proposed IoT-based system in greenhouse agriculture, contributing to increased sustainability, reduced energy consumption, and efficient crop production.

2.1.8 Smart Agriculture Applications

The paper by Oteyo *et al.*[8] emphasizes the importance of Smart Agriculture Applications (SAAs) in modern farming. SAAs help in data collection, processing, and evaluation, overcoming challenges like poor internet connectivity in remote areas.The paper proposes a taxonomy for classifying SAAs based on architectural models, target platforms, and software engineering issues. It highlights the role of SAAs in various agricultural domains, such as irrigation management and pest control.Additionally, the paper discusses open issues and research directions, including peer-to-peer models, scalability, performance evaluation, and security concerns.Overall, it underscores how SAAs enhance smart farming by enabling data-driven decision-making, remote monitoring, and overcoming connectivity challenges in agriculture.

2.2 OVERVIEW AND OUTCOME OF LITERATURE SURVEY

The Comprehensive overview of Literature survey is given in Table 2.1 and the Outcome obtained is given in Table 2.2.

Sl No	Paper	Key Idea	Algorithm/Technique Used
1	Saranya *et al.*[1]	Precision agriculture benefits from DL and IoT, enables smart farming	DL for Pest Detection, Transfer Learning, IoT Architecture, Sensor Classifications.
2	Durai *et al.*[2]	ML and DL app model for crop monitor (prediction, pest and weed identification, cost estimate)	Resnet152V2 Pretrained Algorithm, Random Forest Algorithm, XG Boost Regressor
3	Banu *et al.*[3]	Predicting crop yield with machine learning (Random Forest Algorithm) on environmental factors	Random Forest Algorithm
4	Nguyen *et al.*[4]	Utilizing MobileNetV2 architecture for rapid and accurate detection in outdoor images.	MobileNetV2
5	Narbaev *et al.*[5]	Proposal of XGBoost algorithm for accurate, timely cost forecasting in project management.	XGBoost
6	Unal *et.al*[6]	Advantage of DL in agriculture for object detection (pest, weed)	Convolutional neural network
7	Lanitha *et al.*[7]	Use of IoT and data analysis for smart agriculture by use of crop environment	IoT, Cloud-based architecture, Data analysis, Machine Learning.
8	Oteyo *et al.*[8]	Use of mobile software for tasks (irrigation scheduling, crop monitoring, pest control, farm management)	MobiCrop, AgriMaps, Agro Decisor EFC, etc.

Table 2.1: Overview: Literature Survey

Sl No	Paper	Good Things About the Work	Drawback/Limitations
1	Saranya *et al.*[1]	Comprehensive review of DL applications in agriculture and IoT automation.	Limited details about data sources and dataset used
2	Durai *et al.*[2]	Improve farming efficiency, productivity, better decisions on crop cultivation, pest management.	Requires significant investment in technology
3	Banu *et al.*[3]	High prediction accuracy, User-friendly interface, Application in agriculture, Consideration of climatic parameters	Limited dataset
4	Nguyen *et al.*[4]	Addresses imbalance in outdoor backgrounds, thereby enhancing overall detection performance.	Limited computing resources.
5	Narbaev *et al.*[5]	Improved accuracy in cost forecasting at all project stages.	Limited dataset
6	Unal *et al.*[6]	Improve the accuracy and efficiency (crop yields, reduced costs).	Requirement of training data, computational cost
7	Lanitha *et al.*[7]	Use of IoT, data analysis for smart agriculture by use of crop environment	Data missing issues in IoT
8	Oteyo *et al.*[8]	Mobile application integration provides farmers with easy access and tools for managing their farms.	Security and Privacy

Table 2.2: Outcome:Literature Survey

2.3 CONCLUSION

The comprehensive literature survey presented here delves into the integration of Machine Learning, Deep Learning, and the Internet of Things (IoT) in precision agriculture, elucidating the advancements and influences of various research studies. At the forefront of this transformation is the work of Saranya *et al.*[1], which underscores the transformative potential of deep learning and IoT in precision agriculture. Their research illuminates the diverse applications of these technologies, from precise pest detection using convolutional neural networks (CNNs) to the deployment of IoT sensors for real-time environmental monitoring. By leveraging DL techniques and IoT infrastructure, Saranya *et al.* demonstrate how farmers can make informed decisions based on accurate and timely data, thereby optimizing crop yields and

resource utilization.Building upon this foundation, Durai *et al.*[2] present an integrated approach to precision agriculture, encompassing crop prediction, weed identification, pest recognition, and cost estimation. Their methodology leverages a combination of ML and DL algorithms, such as Random Forest and ResNet152V2, to address various challenges faced by farmers. By harnessing the power of predictive modeling and image recognition, Durai *et al.* empower farmers with actionable insights for crop management and protection, ultimately enhancing productivity and profitability in agriculture.

Banu[3],Devi *et al.*[4], Narbhaev *et al.*[5], and Unal *et al.*[6] contribute diverse algorithmic methodologies to enhance precision agriculture. Banu focuses on predicting crop yields using machine learning algorithms, while Nguyen *et al.*[4] introduce a fresh strategy for object detection, emphasizing its applicability to outdoor images. Narbaev *et al.*[5] discuss the utilization of the XGBoost Regressor for cost forecasting in project management, offering a superior alternative to traditional methods. Unal *et al.*[6] explore the transformative role of deep learning and IoT in agriculture, particularly in smart farming applications.

Furthermore, Lanitha *et al.*[7] and Oteyo *et al.*[8] provide critical insights into the practical implementation of IoT and mobile applications in smart agriculture. Their work emphasizes the importance of standardized architectures, efficient data processing, and user-friendly interfaces in facilitating technology adoption among farmers. By surveying the landscape of IoT-enabled greenhouse agriculture and mobile software applications for farm management, Lanitha *et al.* and Oteyo *et al.* contribute to the ongoing discourse on sustainable farming practices and digital transformation in agriculture.

In conclusion, the integration of Machine Learning, Deep Learning, and IoT holds immense promise for the future of agriculture. These technologies offer solutions for data-driven decision-making, remote monitoring, and overcoming connectivity challenges in agricultural processes.

CHAPTER 3

PROBLEM DEFINITION

3.1 PROBLEM STATEMENT

To develop a Mobile App for a smart farming system that effectively manages agriculture

3.2 PROBLEM DESCRIPTION

Current farming practices lack efficiency and precision, leading to suboptimal outcomes and increased costs. Manual processes and outdated methods hinder optimal resource utilization and crop yield predictions. The absence of advanced technologies like IoT, machine learning, and deep learning exacerbates these challenges. The key issues include manual farming tasks, imprecise crop yield predictions, and inadequate weed and pest management. Existing systems lack a centralized, user-friendly platform for farmers. This project addresses these challenges with a Mobile App for a smart farming system. Objectives include automating tasks with an IoT-enabled smart Greenhouse, implementing machine learning for crop yield and cost estimation, and using deep learning for weed and pest detection. Integrating these modules into a user-friendly mobile app aims to enhance farming efficiency and sustainability, ushering in a new era of technology-driven agriculture.

3.3 SCOPE AND OBJECTIVES

3.3.1 SCOPE

Addressing food insecurity and improving small-scale farm productivity necessitates the adoption of modern, eco-friendly technologies capable of efficiently managing resources for sustainable agriculture. Smart farming is the key, ushering in a future characterized by cost reduction, heightened productivity, and the production of superior-quality agricultural goods. The scope of the FarmAlley app, integrated with IoT for greenhouse automation, is to provide a comprehensive agricultural support system for farmers. This multifaceted application employs machine learning and deep learning algorithms for crop prediction, weed and pest identification, and cost estimation. Additionally, it enables remote greenhouse management through IoT-connected sensors, allowing for automatic control of functions such as sliding windows, cooling fans, irrigation, and lighting. This software aims to optimize crop cultivation, reduce resource usage, and improve farming outcomes, with a strong focus on data-driven decision-making for farmers. It is important to emphasize that the software will not involve agricultural trading or physical crop harvesting, concentrating on aiding farmers in their crop management practices.

3.3.2 OBJECTIVES

The specific objectives of the system are :

1. Automate farming tasks using IoT-enabled smart Greenhouse.
2. Create a machine learning model for crop prediction.
3. Utilize machine learning to estimate agricultural costs.
4. Implement deep learning algorithms for precise weed detection.
5. Develop deep learning models to detect pests.
6. Integrate all the modules inside mobile app and enable notifications.

CHAPTER 4

REQUIREMENT SPECIFICATION

4.1 REQUIREMENT SPECIFICATION

In this chapter, the requirement specification of the project is outlined. It delineates crucial functionalities essential for optimizing agricultural management in Smart Farming using IoT, DL, and ML. This foundational document systematically gathers stakeholder needs, guiding the development of a user-friendly mobile application. Through this application, farmers gain the capability to monitor and control various aspects of their agricultural operations, including environmental control, crop prediction, weed and pest detection, and cost estimation. Non-functional requirements ensure system reliability, scalability, and performance, while interfaces such as IoT sensors and cloud servers facilitate seamless data exchange. Clear acceptance criteria validate system readiness, ensuring alignment with stakeholder expectations and driving the success of smart farming initiatives.

4.2 SOFTWARE REQUIREMENT SPECIFICATION

4.2.1 Introduction

Software Requirement Specification (SRS) provides an overview of the entire SRS with purpose, scope,definitions, acronyms, abbreviations, references and overview of the requirements. The aim of this document is to gather and analyze and give an in-depth insight of the Smart Farming Using Machine Learning, Deep Learning and IoT by defining the problem statement in detail. Nevertheless, it also concentrates on the capabilities required by stakeholders and their needs while defining high-level product features.

4.2.2 Purpose

The purpose of the requirement specification is to collect and analyze all assorted ideas that have come up to define the system, and its requirements with respect to the users of the system. Also, we shall predict and sort out how we hope this product will be used to gain a better understanding of the project, outline concepts that may be developed later, and document ideas that are being considered, but may be discarded as the system develops. The system is a app developed for advanced tech controlled smart farming system using techniques such as IoT,Machine Learning and Deep Learning. The system leverages various datas collected from the farming environment to inspect, recommend and control systems. We aim to improve crop selection, reduce resource consumption, and enable proactive management of environmental conditions by harnessing cutting-edge technologies. This will ultimately lead to sustainable and profitable farming practices.

4.2.3 Product / System Overview

The Agricultural Automation System is designed to revolutionize farming by employing advanced technology and automation. This section outlines the primary functions and capabilities of the software.

1. **User-Friendly Mobile Application**: A mobile application that serves as an

intuitive interface for farmers to monitor and control various aspects of their agricultural operations. This user-friendly app enhances accessibility and simplifies the management of automated systems.

2. **Environmental Control**: Precise regulation of environmental conditions within the farming facility, including temperature, humidity. This control ensures an ideal climate for optimal plant growth, ultimately leading to higher yields and healthier crops.
3. **Cooling Fan Controller**: The Cooling Fan Controller regulates the operation of cooling fans within the farming environment. It employs temperature sensors and control algorithms to maintain optimal climate conditions for plant growth.
4. **Irrigation Management** : Automated irrigation control based on real-time soil moisture data. It interfaces with moisture sensors and water distribution systems to ensure plants receive adequate hydration.
5. **Smart Lighting Solutions**: Implementation and management of artificial lighting systems that simulate natural light cycles. This feature promotes plant development, flowering, and fruiting, leading to higher crop yields.
6. **Sliding Window Controller**: The Sliding Window Controller automates the opening and closing of windows to regulate airflow and temperature within the farming facility. It interfaces with motorized window mechanisms and environmental sensors.
7. **Crop Prediction**: Leveraging geographical data and machine learning algorithms to forecast which crops are best suited for a specific plot of land. Factors like soil type, climate conditions, and historical data are considered to provide valuable recommendations to farmers.
8. **Weed and Pest Detection**: Utilizing image classification techniques to identify and categorize weeds and pests in the farming area.This information empowers farmers to implement targeted and effective pest and weed control measures.
9. **Cost Estimation**: This module analyzes historical cultivation data of India spanning from 2009 to 2021. It forecasts the cost estimation of cultivation for

various crops across different states, considering the specified land in hectares. Its predictive capabilities extend up to 2031, providing insights for the next decade, comprising seven years from the present year.

The block diagram showing the different functions/components is as shown below in Figure 4.1. The detailed view of each module is provided in Figure 4.2, 4.3, 4.4, 4.5, 4.6.

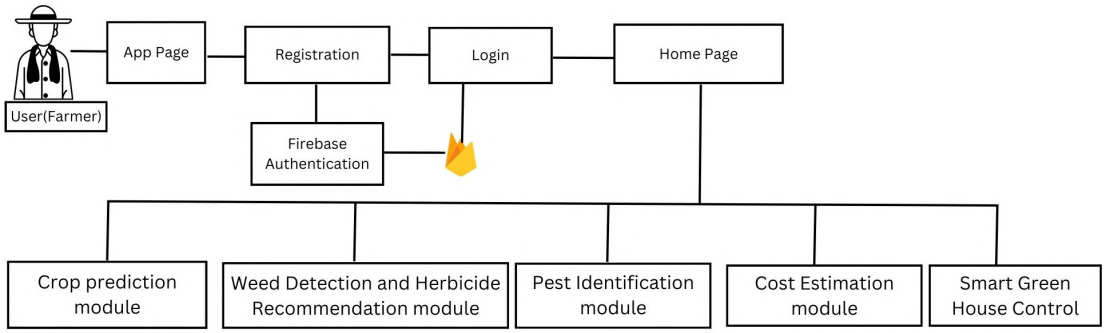


Figure 4.1: System Overview.

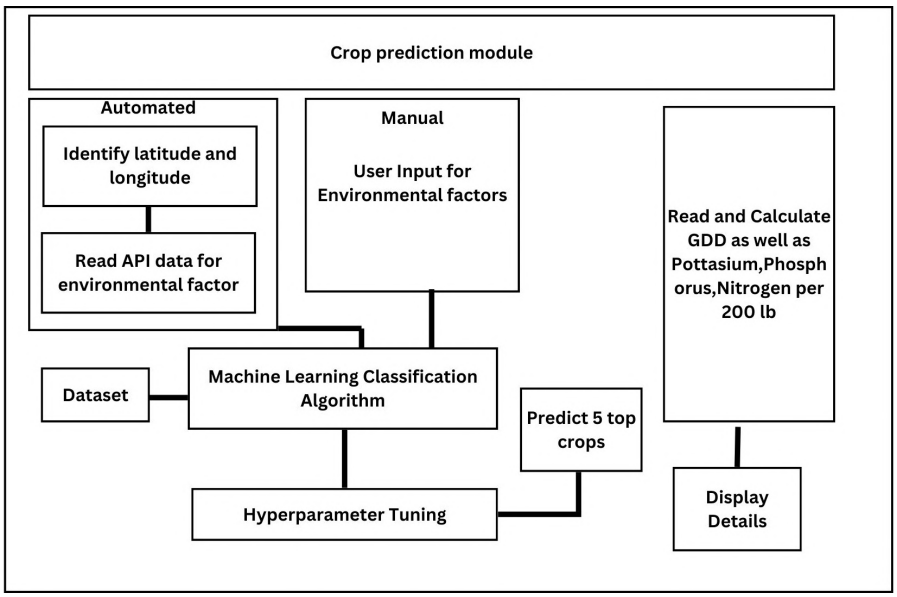


Figure 4.2: Crop Prediction Module.

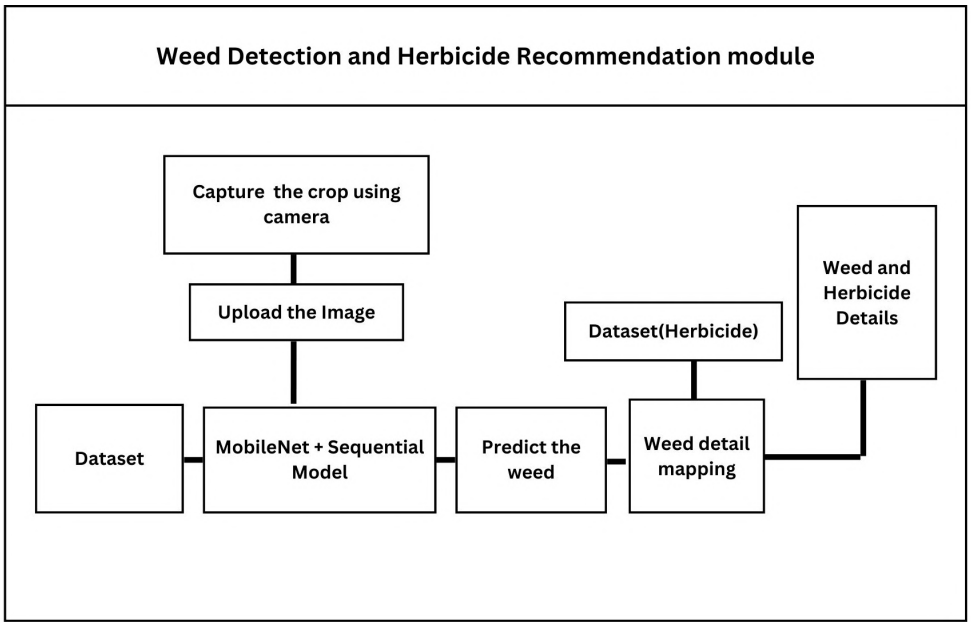


Figure 4.3: Weed Identification Module.

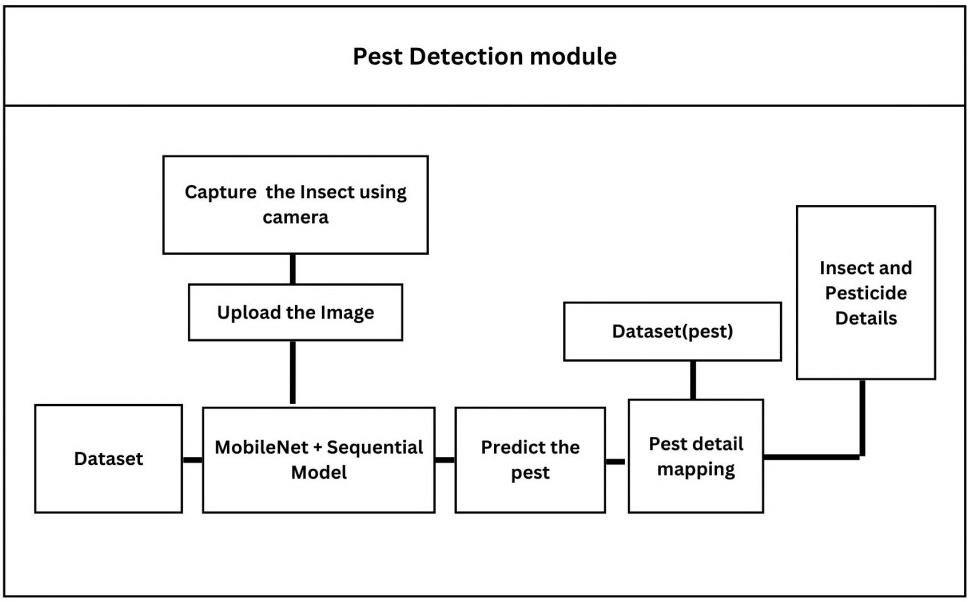


Figure 4.4: Pest Identification Module.

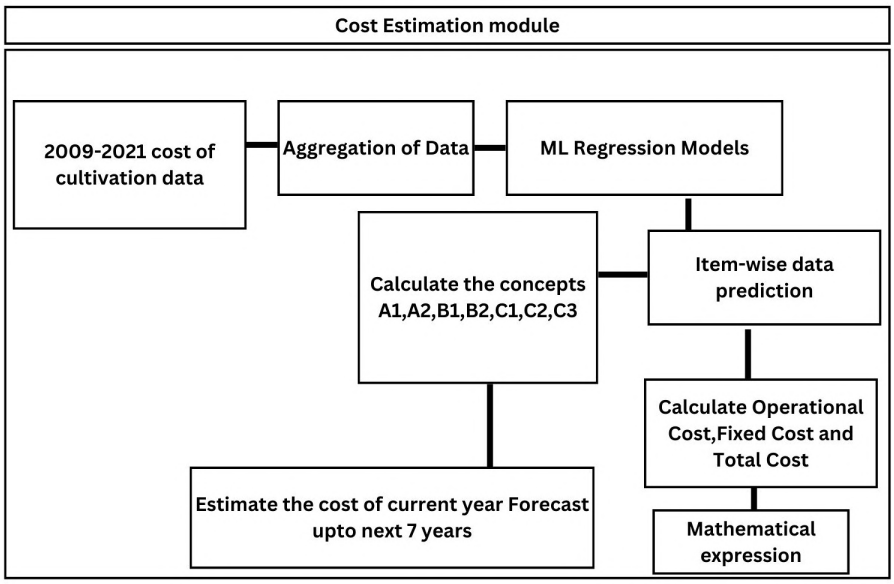


Figure 4.5: Cost Estimation Module.

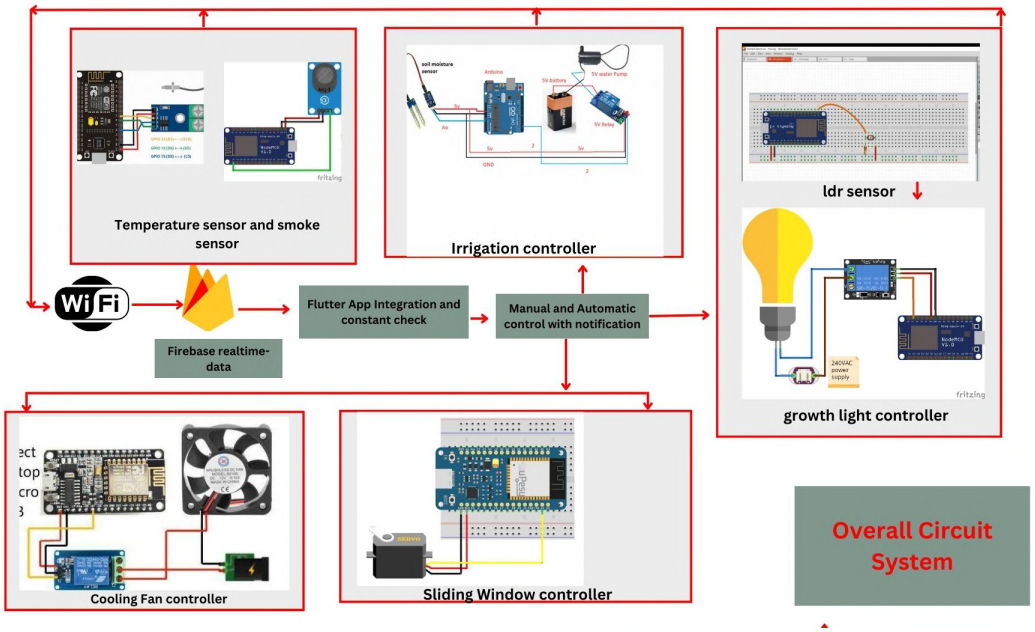


Figure 4.6: IoT Circuit System.

4.2.4 User Characteristics

There are two types of users that interact with the system:

1. Non-technical public people who interact with the system. Non-technical people in this context refer to the farmers who interact with the system. The farmer should have basic knowledge about the usual registration, login system, and be familiar with the app interfaces. They should also know how to read and understand English and be familiar with agricultural factors such as pH, herbicide, and growth lights.

2. Administrators who manage the system. The administrator can manage the interface and shall be able to make updates of software as well as the data used for training the system that affect the agricultural factors and thus the output.

4.2.5 Constraints

Constraints are the things that control the development of the system by restricting the operation of the system under specific and well-defined bounds. The constraints regarding the system are related to predictive action, network reliability and cost-effectiveness.

1. **Sensor costs**

   The crop prediction make use of the factors such as pH value which have to be entered manually as the pH sensors for testing are heavily priced and is an unnecessary addition as the pH of the soil can be easily and feasibly tested from the nearby soil testing lab.The methodology also uses nutrient values which are feeded according to the geographical location or either enterd by farmers and are not tested manually by sensors as it is also heavily priced.

2. **Automations**

   The Project purely relies on monitoring and automating further actions like herbicide spraying, Pest removal, tilling, harvesting etc. have to be performed by the farmer and the robotic actions are not covered under this project due to limitation of resources, time and workspace to act upon.

3. **Network and Power supply**

   The System requires strong internet connections with stable GPS access and a reliable power supply . The internet connections is required for sending signals from the microprocessor systems in the system , fetching output of ML and DL models to mobile application, and power source is required for powering the IoT control in DC(Direct Current)/AC(Alternating Current) power source.

4. **Decoupling ML and DL from IoT Control**

   The Crop Prediction, Cost Estimation, Weed Identification, and Pest Identification modules are designed for crops grown in larger farm fields. These modules utilize machine learning and deep learning techniques to support farmers in managing crops efficiently across vast agricultural areas. On the other hand, the IoT Control System is tailored specifically for managing greenhouse environments. It focuses on controlling and optimizing environmental conditions within greenhouses, such as temperature and humidity, to ensure optimal growth. While the greenhouse IoT Control System uses some common parameters such as temperature and humidity, these values are not directly used for machine learning and deep learning models in the other modules but is given as a choice. This separation allows farmers growing crops in larger areas to benefit from specialized models without being constrained by the specific requirements of greenhouse farming. As a result, farmers can leverage these systems for a variety of agricultural purposes throughout the year.

4.2.6 Assumptions and Dependencies

1. **Interfacing of IoT and App**

   The IoT microprocessors are meant to communicate to the Flutter app via Firebase using the real-time data via Wi-Fi communication, however since the response system should be strong from the app to the circuit and have multiple circuits the reliability on firebase may be subjected to change over time.

2. **Circuit failure**

   The project makes use of microprocessors, sensors and other various components as well as power source, as any other circuit system, this system is too vulnerable to short circuit or damage of the components and may require the replacement with other components for enhanced performance overtime. The circuit is used for building of IoT control related to various activities like cooling fan controller, Sliding window controller, Irrigation system, growth light control.

3. **Constrain on Mobile App Deployment**

   The project aims to build a mobile app using Flutter however if the response interface of IoT and Flutter faces obstruction during mobile app deployment, the system shall be deployed in mobile app

4.2.7 Functional Requirements

The system should be capable of monitoring the agriculture environment, then can analyze the current situation and can control accordingly.

1. **Crop Prediction**

   a) The application should have a crop prediction module that suggests suitable crops based on the specific land's characteristics.

   b) Users should be able to input either location's data or environmental factors such as N,P,K,pH,Rainfall,Temperature,Humidity etc. and the system should provide a list of recommended crops.

Weed Identification

2. a) The application should feature a weed identification system using image classification.

   b) Users should be able to take or upload images captured from the field along crop name. The Model predicts the weed name along with a list of herbicides followed by its dosage and other details.

3. **Pest Identification**

   a) The application should provide a pest identification module using image classification

   b) Users should be able to take or upload images captured from the field along crop name and the model predicts the pesticides suitable for the pest.

4. **Cost Estimation**

   a) The application should include a cost estimation module for predicting the minimum cost of crop cultivation.

   b) Users should be able to select the crop and location, the model predicts the Cost of Cultivation details till 2031

5. **Plant Growth Light System**

   a) The application should enable users to control the automated plant growth light system for optimized plant growth manually as well as automated control by environmental factors.

   b) Grow Light Bulb is used for customizing light schedules and intensity levels based on plant requirements.

6. **Irrigation Controller**

   a) The application should provide an irrigation control feature that allows users to schedule and manage irrigation activities.

   b) Soil moisture sensor is used for detecting the moisture content in soil and controlling the motor pump and control manually as well as automated control by environmental factors.

7. **Sliding Window Controller**

   a) The application should allow users to remotely control sliding windows for ventilation and temperature regulation.

   b) Users should be able to set window opening and closing schedules manually as well as automated control by environmental factors.

8. **Cooling Fan Controller**

   a) The application should allow farmers to control cooling fans remotely through the mobile interface.

   b) Users should be able to set on and off manually as well as automated control by environmental factors.

4.2.8 Non-Functional Requirements

1. **Performance Requirements**
   a) **Task Processing time frame**: The software must process automatic greenhouse control tasks, such as window and fan adjustments, irrigation, and lighting changes, within an acceptable time frame from user input.

   b) **Response Time**: User interactions and data transactions, including image analysis for weed and pest identification, should be completed promptly to ensure a responsive user experience.

   c) **Information Handling**: The application should effectively manage agricultural data for multiple farms,including crop data, sensor readings, and user profiles.

2. **Software System Attributes**

   a) **Usability**: The system should be easy to use with clear instructions and simple controls and regardless of the technical skills, find it easy to operate.

   b) **Scalability**: The system should grow with the user's needs, allowing them to add or remove components as necessary.

   c) **Reliability**: The system should work reliably and recover gracefully from errors.

   d) **Availability**: The software should be accessible whenever users need it.

   e) **Maintainability**: Updates and maintenance should be straightforward and well-documented.

4.2.9 External Interface Requirements

Users interact with the smart farming system through various interfaces and devices. These interfaces and devices provide access to and control over the system's features and data, ensuring efficient interaction with the technology.

1. **User Interfaces**

   The various instances of the interface are shown in Figure 4.7,4.8,4.9,4.10,4.12, 4.13,4.14 respectively. The given designs are representative samples only and they may vary in their actual design and layout depending on the implementation specifics.

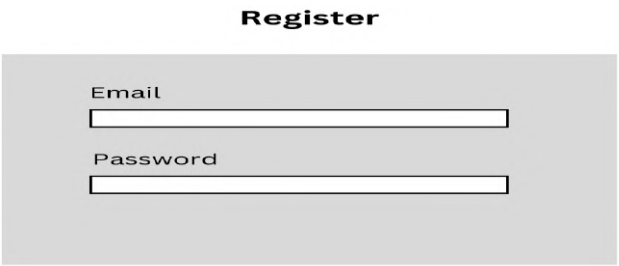


Figure 4.7: Registration Page

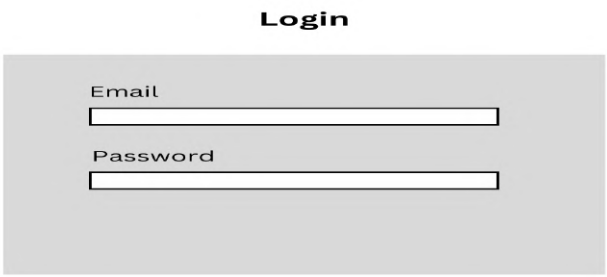


Figure 4.8: Login Page

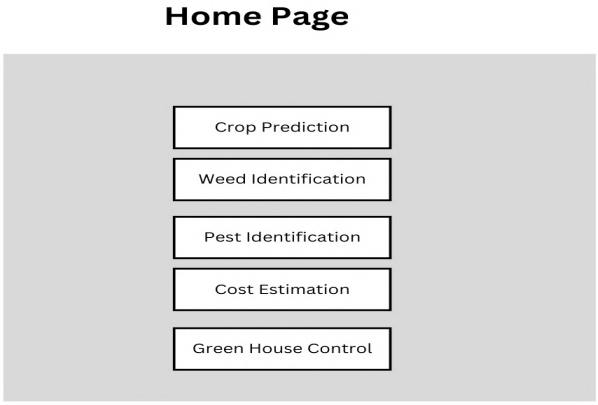


Figure 4.9: Home Page

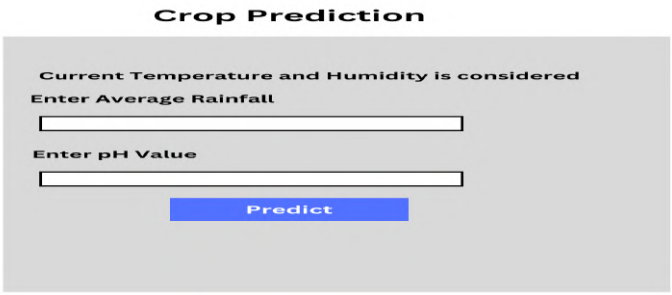


Figure 4.10: Crop Prediction Page

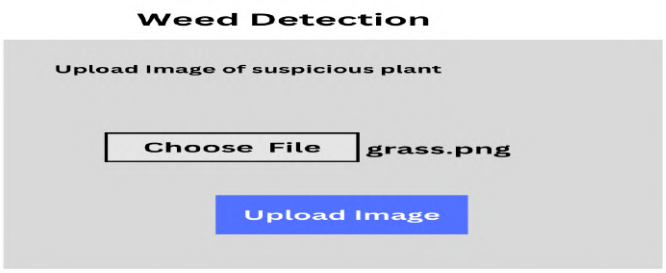


Figure 4.11: Weed Identification Page

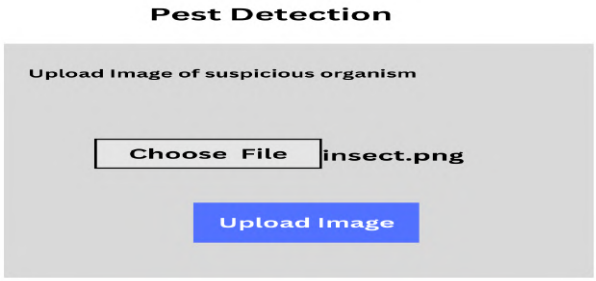


Figure 4.12: Pest Identification Page

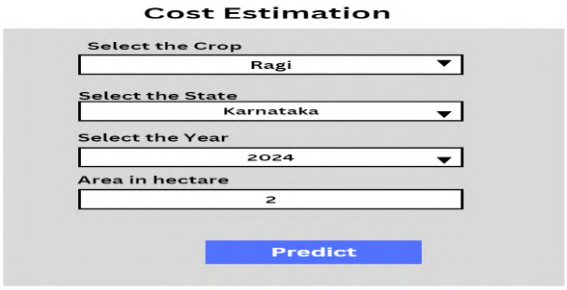


Figure 4.13: Cost Estimation Page

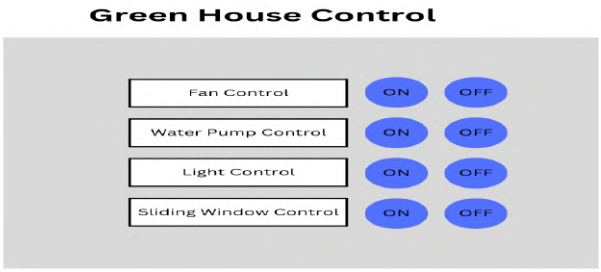


Figure 4.14: Greehouse Control Page

2. **Hardware Interfaces**

The Hardware Interfaces and associated communication protocols are represented in Table 4.1.

Hardware Interfaces	**Supported Devices/Components**	**Communication Protocols**
IoT Interfaces	• Temperature sensor • Soil Moisture Sensor • Smoke Sensor • Light Intensity Sensor	Wi-Fi (analyzed in ThinkSpeak Platform) through Inbuilt Wi-Fi module in NodeMCU
Greenhouse Automation	• Window and fan automation controller • Lighting control systems • Irrigation systems	Wi-Fi (response action via Firebase) through Inbuilt Wi-Fi module in NodeMCU
Wireless Connectivity	Device should support 3G, GPRS, and Wi-Fi 802.11 b/g/n connectivity	Nil
Cloud Servers (Firebase)	Firebase cloud servers for data storage and user authentication	HTTPS

Table 4.1: Hardware Interfaces and Communication Protocols

3. **Software Interfaces**

The Software Interfaces along with Name,Version,Source are represented in Table 4.2,each of the Interface definition and purpose is given in Table 4.3.

Software Interfaces	Name	Version	Source
Data Management Systems	Firebase (Realtime Database, Firestore)	Latest	Firebase Console
Operating System	Android (for mobile app)	Latest	Android Open-Source Project (AOSP)
Other Required Software	Flutter (UI framework)	Latest	Open-source community

Table 4.2: Software Interfaces

Interfaces	Purpose	Interface Definition
Data Management System	Firebase is used for real-time data synchronization, authentication, and cloud storage in the app.	Firebase provides documented APIs and SDKs that define the structure and format of data messages.
Operating System	The app is designed for mobile devices running Android, so it interfaces with the respective operating systems for platform-specific functionalities.	The interfaces with Android are well-documented by the Android Open-Source Project.
Other Required Software	Flutter is the primary software used for developing the app's user interface and providing a cross-platform framework.	The interface with Flutter is defined by the Flutter SDK and its documented APIs.

Table 4.3: Interfaces, Purpose, and Interface Definitions

4. **Communication Interfaces**

   - HTTPS protocol shall be used for communication over the Internet.
   - WPA (Wi-Fi Protected Access) or WPA2 supported devices must be used.

4.2.10 Acceptance Criteria

The various criteria for product acceptance are listed below:

1. Does the system meet agricultural requirements?
   YES/NO

2. Is data accuracy and quality assured?
   YES/NO

3. Is the system user-friendly?
   YES/NO

4. How does the system handle environmental conditions?
   a. Very Fast, b. Fast, c. Moderate, d. Slow, e. Very Slow

5. How are the energy and resource requirements?
   a. Very Fast, b. Fast, c. Moderate, d. Slow, e. Very Slow

6. Does the system offer predictive capabilities?
   YES/NO

7. How secure and resilient is the system?
   a. Very Fast, b. Fast, c. Moderate, d. Slow, e. Very Slow

8. Does the system provide scalability and integration options?
   YES/NO

CHAPTER 5

PROJECT DESIGN

In this chapter, the design process for our Smart Farming project is outlined, emphasizing the utilization of various design models such as UML, use-case diagrams, design models, state transition diagrams, sequence diagrams, and entity-relationship (ER) designs. These models play a pivotal role in conceptualizing, visualizing, and communicating the architecture and functionality of our smart farming system, integrating IoT, DL, and ML technologies. By exploring the application of each design model within the context of smart farming, we aim to elucidate their importance in creating efficient and sustainable agricultural solutions. Through various insights, we highlight the critical role of effective project design in driving the success of our smart farming initiative.

5.1 SOFTWARE DESIGN DOCUMENT

5.1.1 Introduction

A software design is a representation of a software designs that are to be used for recording design information, addressing various design concerns, and communicating that information to the design's stakeholders. This document accompanies an architecture diagram with pointers to detailed feature specifications of smaller pieces of the design. Practically, the description is required to coordinate a large team under a single vision, needs to be a stable reference, and outlines all parts of the software and how they will work.

5.1.2 Purpose

The Software Design document serves as a blueprint for the development of an innovative farming assistance application leveraging Machine Learning (ML), Deep Learning (DL), and the Internet of Things (IoT). Its primary objective is to provide a comprehensive description of the system's design, enabling software development to proceed with clarity regarding what needs to be built and how it should be constructed. It delineates the integration of cutting-edge technologies such as ML, DL, and IoT to empower farmers with advanced technological assistance in various aspects of farming. From crop prediction to pest management and environmental monitoring, the system aims to revolutionize traditional farming practices by harnessing the power of data-driven decision-making. By formalizing the design of the application, this document facilitates collaboration among developers, stakeholders, and end-users, fostering a common understanding of the system's objectives and functionalities. Ultimately, it lays the foundation for the successful development and implementation of the farming assistance application, poised to significantly enhance agricultural productivity and sustainability.

5.1.3 High Level Design (System Architecture)

The proposed system facilitates user authentication, enabling users to sign up and log in securely. Upon authentication, users gain access to the user profile management and dashboard module, where they can manage their account settings and view personalized farming predictions generated by ML/DL predictors on the homepage. Sensor data from agricultural environments is seamlessly ingested through the IoT integration component, providing real-time insights for informed decision-making.Meanwhile, the admin, equipped with login credentials, can access various modules and the database to modify and update user details as needed. The use case diagram encapsulates these interactions, illustrating the functional relationships between actors (users and admin) and system components (authentication, user profile management, dashboard module, IoT integration, and data storage).The use case diagram is shown in Figure 5.1

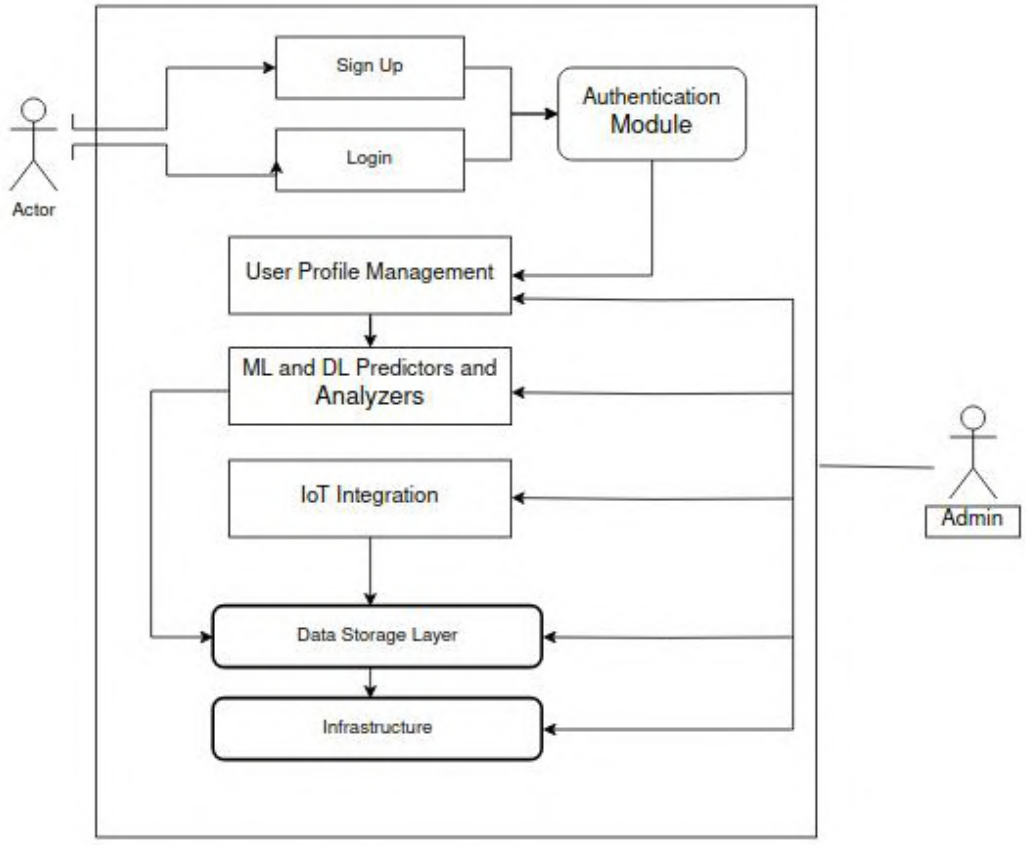


Figure 5.1: Use Case Diagram

A deployment diagram in the Unified Modelling Language models the physical deployment of artifacts on nodes. Figure 5.2 illustrates the deployment diagram of the proposed system

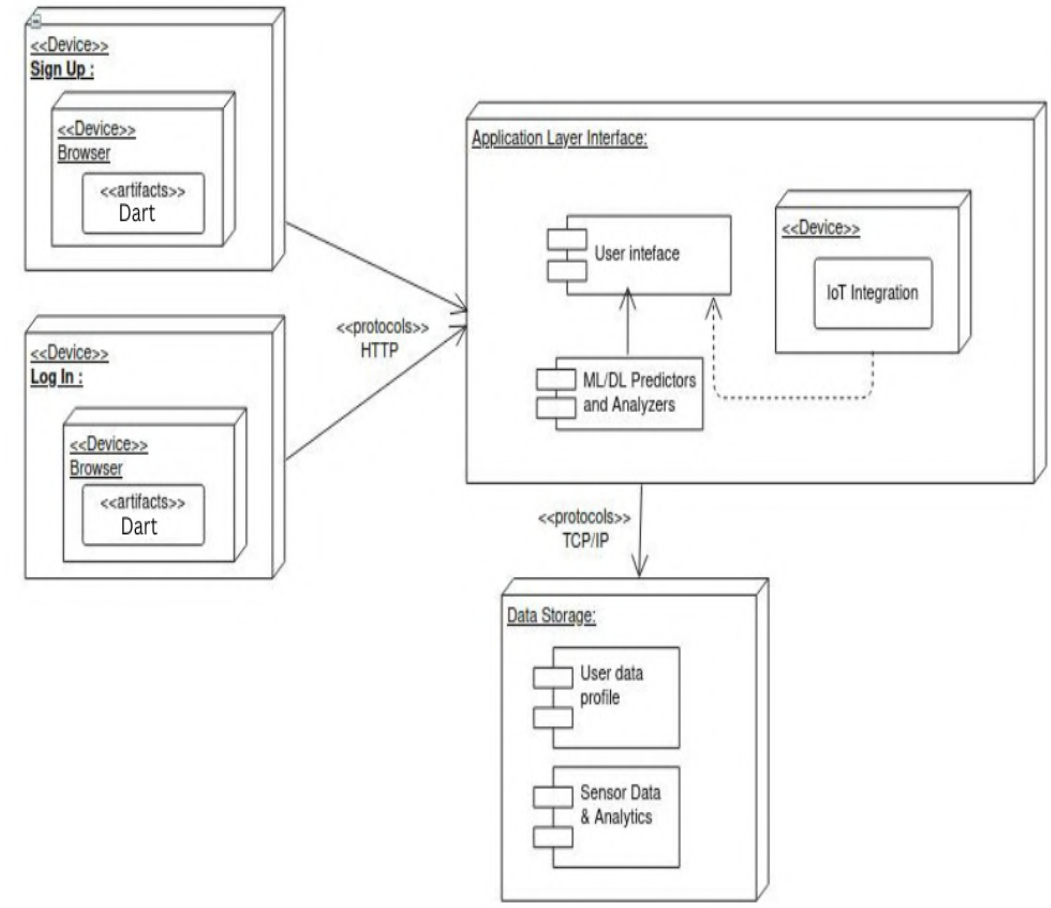


Figure 5.2: UML Deployment Diagram

5.1.4 Low Level Design (Detail Design)

1. **Analysis Model**

   The sequence diagrams depict interactions between objects in sequential order, illustrating the flow of interactions within the system. Figure 5.3 presents the sequence diagram for the ML and DL components, while Figure 5.4 displays the sequence diagram for the IoT component. Users access the interface using their credentials and interact with various modules such as crop prediction, cost estimation, weed identification, pest detection, and IoT control. Responses from each module are traced back to the user. Although represented as separate figures due to visibility constraints, in reality, they are part of a single system.

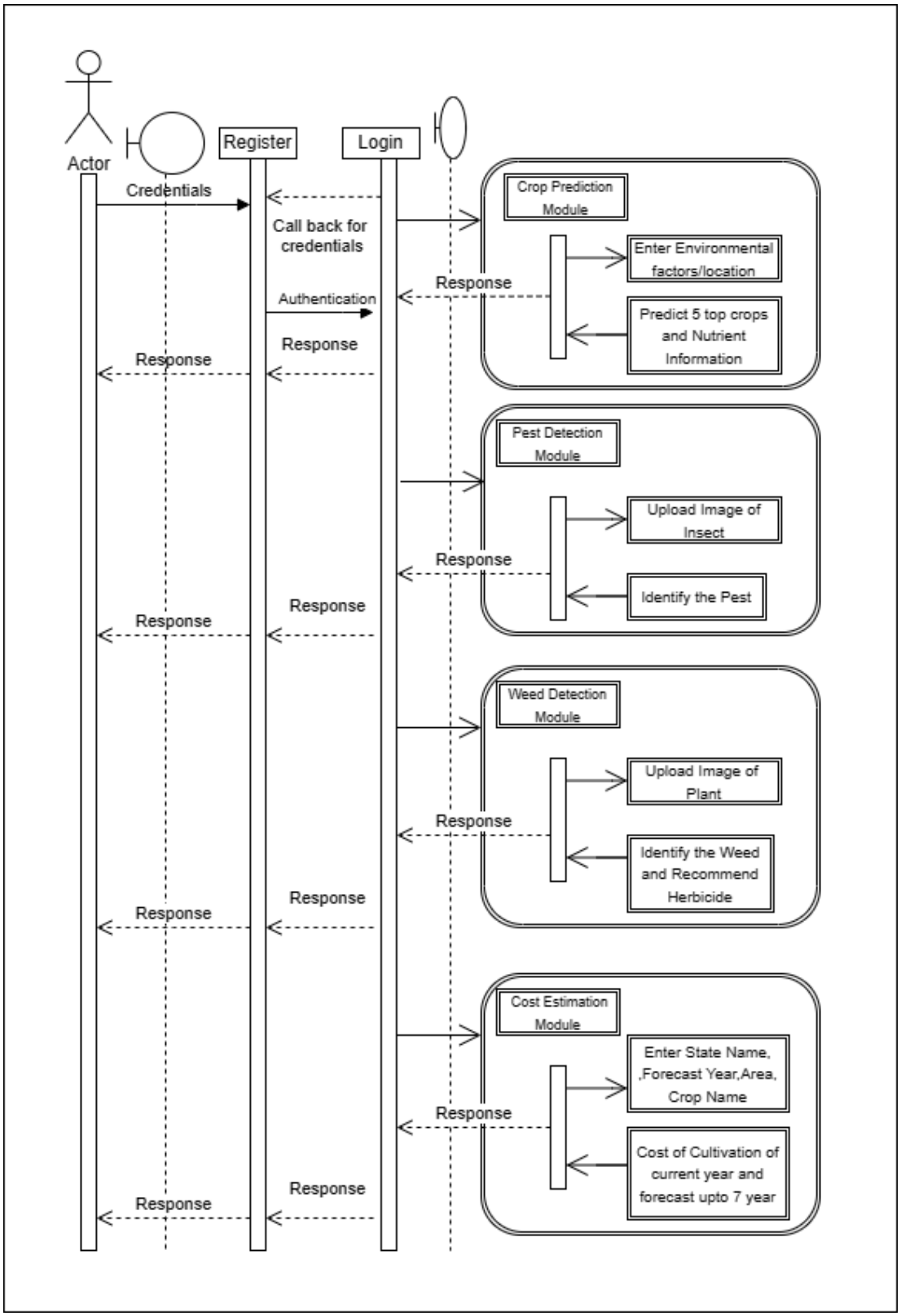


Figure 5.3: Sequence diagram .

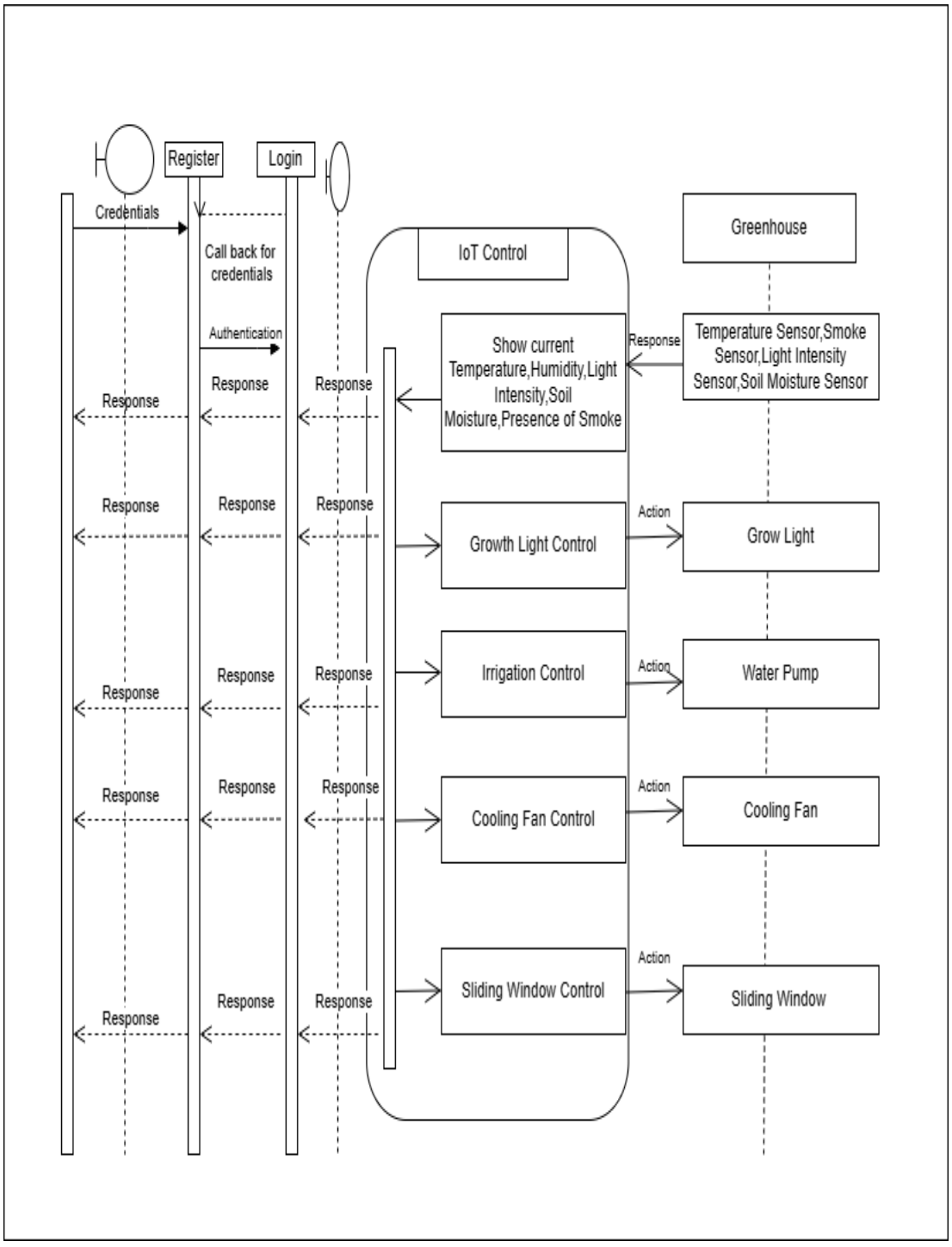


Figure 5.4: Sequence diagram.

2. **Design Model**

State transition diagram is a type of diagram that is used to represent different transition (changing) states of a System.The state transition diagram of the system for ML and DL model is shown in Figure 5.5 The state transition diagram of the system for IoT model is shown in Figure 5.6 and The user can access the system and various authentication are provided by which invalid values shall be rejected in login and incorrect values shall not be considered for factors such as Crop prediction,Cost estimation,Image deprecation in Pest or Weed Identification. (NB:The Figure 5.5 and Figure 5.6 forms part of a single system in reality , however due to the constraint of representing large image with visibility inside the document,the system is shown as two different figures).

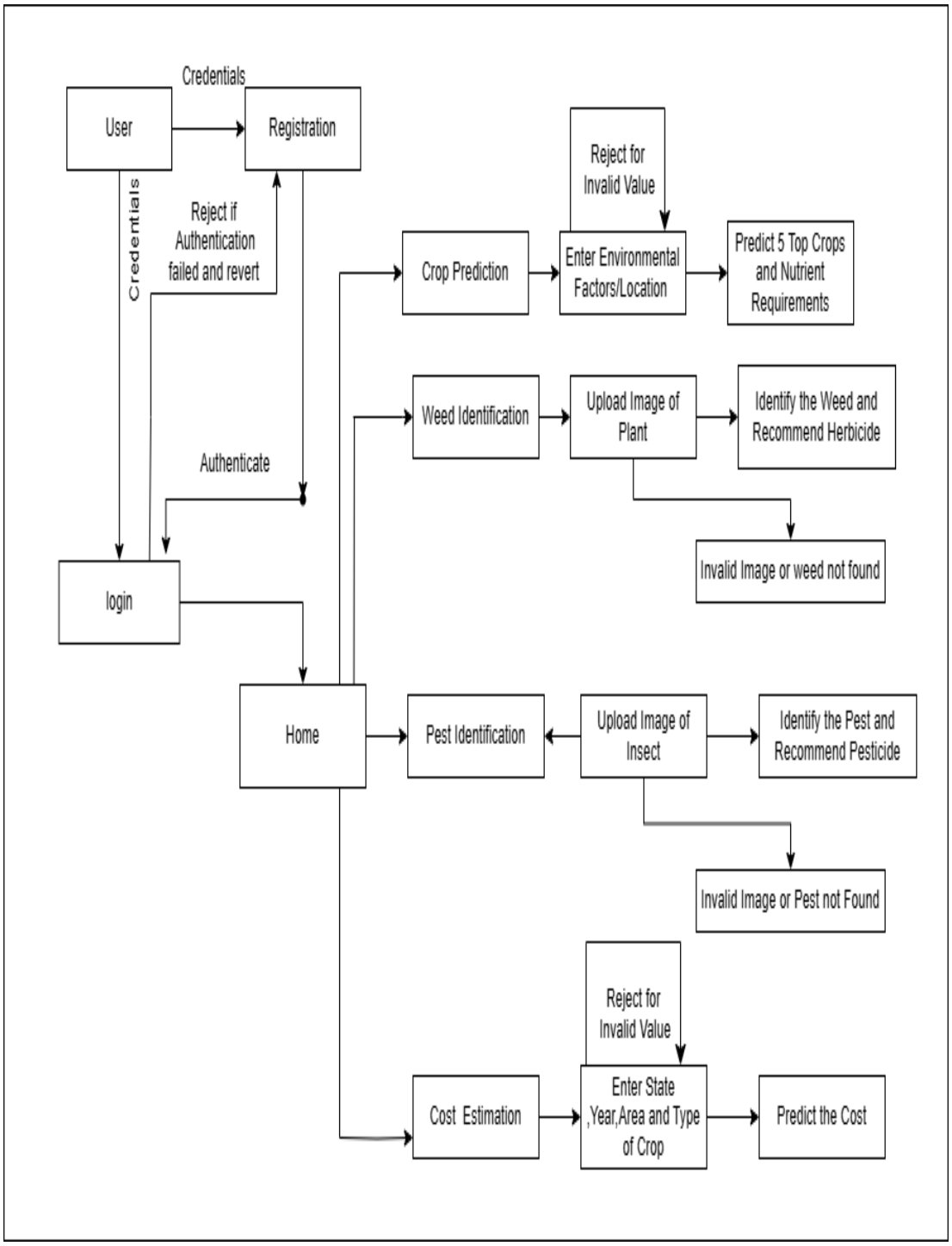


Figure 5.5: State transition diagram

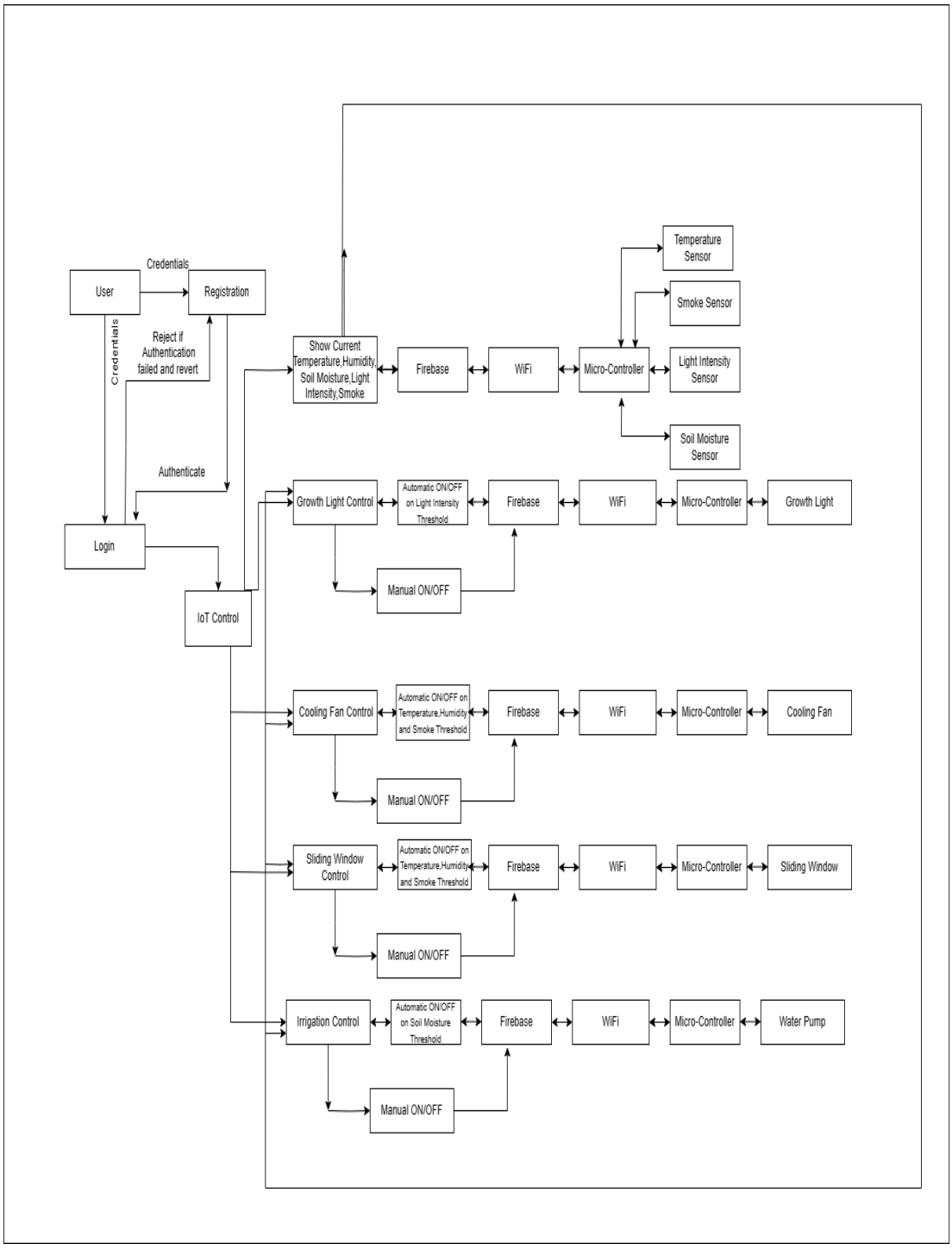


Figure 5.6: State transition diagram

5.1.5 Database Design

ER Model

The following shows the ER table description(Table 5.1)and ER Model Diagram of the NoSQL database related to each functionality of various modules in app is shown in Figure 5.7.

User	**Function**	**Key**	**Extra**
Uu_id- email	Integer	Primary key	Unique
Name	String		
Email	String		Unique
Password	Integer		

Table 5.1: Field Description

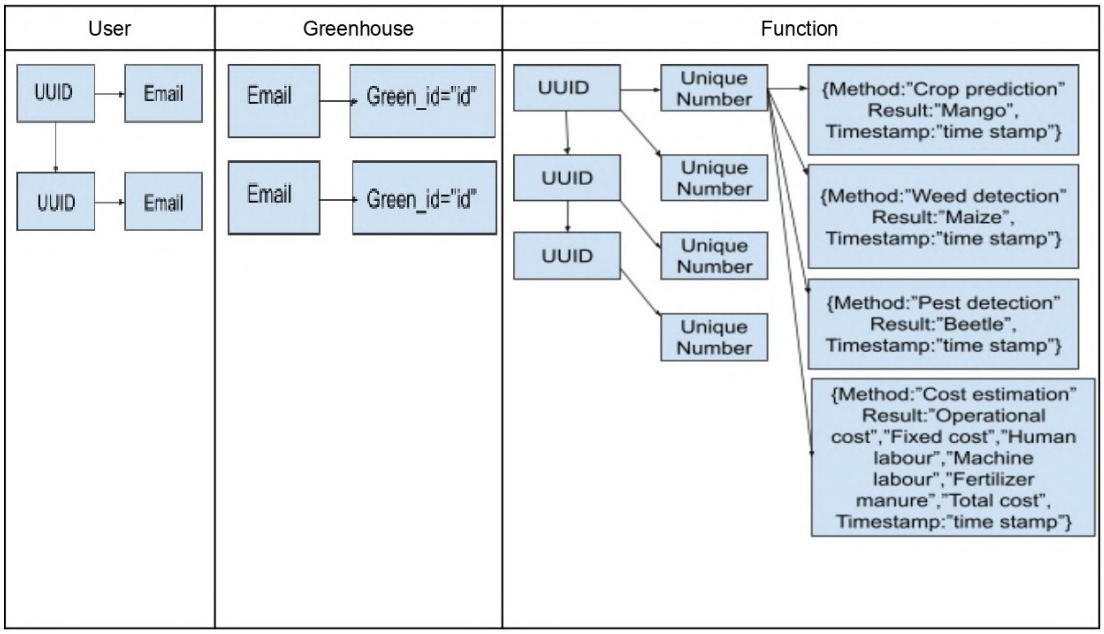


Figure 5.7: ER Diagram

CHAPTER 6

IMPLEMENTATION

6.1 IMPLEMENTATION TOOLS

6.1.1 Mobile App Development Tools

1. **Flutter**

   Flutter is a UI toolkit developed by Google for building natively compiled applications for mobile, web, and desktop from a single codebase. It provides a framework for building user interfaces using a reactive programming paradigm and a rich set of pre-built UI components.

2. **Firebase**

   Firebase is a platform developed by Google that provides a variety of services and tools to help developers build, improve, and grow their applications. It offers services such as real-time database, authentication, cloud storage, hosting, analytics, and more, which can be integrated into mobile and web applications to add functionality and improve user experience.

3. **Android Studio**

   Android Studio is the official integrated development environment (IDE) for Android app development, provided by Google. It offers tools for writing, debugging, and testing Android applications, as well as features for managing project files, dependencies, and resources.

6.1.2 Backend Development Tools

1. **Python**

   Python is a versatile and widely-used programming language known for its simplicity and readability. It offers a rich ecosystem of libraries and frameworks that make it suitable for various types of software development, including web development.

2. **Flask**

   Flask is a lightweight and flexible web framework for Python. It provides developers with the tools and libraries needed to build web applications quickly and efficiently. Flask is known for its simplicity, minimalism, and extensibility, making it an excellent choice for developing small to medium-sized web applications.

3. **Jupyter**

   Jupyter is an open-source web application that allows you to create and share documents containing live code, equations, visualizations, and narrative text. It supports various programming languages, but it's particularly popular in the Python community for interactive data analysis and scientific computing.

4. **Google Colab**

   Google Colab, short for Colaboratory, is a cloud-based Jupyter notebook environment provided by Google. It allows you to write and execute Python code in a browser, with access to free Graphic Processing Unit(GPU) and Tensor Processing Unit(TPU) resources. It's commonly used for machine learning experiments, data analysis, and collaborative coding.

5. **Python Anywhere**

   Python Anywhere is a cloud-based Python development platform that provides a web-based Python environment for running Python scripts, web applications, and scheduled tasks. It offers features like web hosting, database access, and scheduled tasks, making it suitable for deploying and hosting Python applications online.

6.1.3 Hardware Components and Modules

1. **ESP8266**

   The ESP8266 is a versatile Wi-Fi module commonly used in IoT projects. It allows devices to connect to the internet and communicate wireless, making it ideal for applications such as home automation, sensor monitoring, and remote control.

2. **MQ2**

   The MQ2 is a gas sensor capable of detecting various gases such as methane, propane, and carbon monoxide. It is commonly used in air quality monitoring systems, gas leakage detection devices, and safety alarms to detect the presence of hazardous gases.

3. **MAX6675**

   The MAX6675 is a thermocouple-to-digital converter module used for measuring temperature. It can accurately read the temperature of a wide range of materials and is commonly used in industrial processes, temperature monitoring systems, and temperature-controlled appliances.

4. **LDR**

   LDR(Light Dependent Resistor)s are light-sensitive resistors that change resistance based on the intensity of light falling on them. They are often used in light sensing circuits, automatic streetlights, and camera exposure control systems to adjust device behavior based on ambient light conditions.

5. **Arduino IDE**

   The Arduino Integrated Development Environment (IDE) is a software platform used for programming and uploading code to Arduino micro-controllers. It provides a user-friendly interface for writing, compiling, and uploading code, making it accessible to beginners and experienced developers alike.

6. **Grow Light**

   Grow lights are artificial light sources designed to stimulate plant growth by emitting light suitable for photosynthesis. They are commonly used in indoor gardening, hydroponic systems, and greenhouse environments to provide supplemental or primary light to plants.

7. **CPU Cooling Fan**

   CPU(Central Processing Unit) cooling fans are used to dissipate heat generated by computer processors to prevent overheating and maintain optimal operating temperatures. They are essential components in desktop computers, laptops, and other electronic devices with high-performance CPUs.

8. **Mini submersible water pumps**

   Mini submersible water pumps are compact pumps designed to operate underwater. They are commonly used in aquariums, fountain displays, hydroponic systems, and water cooling setups to circulate water and maintain proper water flow and circulation.

9. **Servo Motor**

   Servo motors are rotary actuators that precisely control angular position, velocity, and acceleration. They are commonly used in robotics, remote control systems, and automation applications to move and control mechanical components with precision and accuracy.

6.2 RESULTS AND DISCUSSION

The results are depicted in this section. We developed a mobile application interface and a greenhouse prototype.

6.2.1 Authentication

The app consists of an opening App Page leading to the Sign-Up Page, where data is stored in Firebase. This is followed by the Login Page, which validates the presence of an existing user. The interfaces are as shown in Figure 6.1.

6.2.2 Dashboards

The app features four main dashboards. The first is displayed on the Home Page to access different service modules. Additionally, a dashboard page is provided for Crop Prediction, which allows the user to choose between Manual, Automated, or Partially Automated services. The Greenhouse Dashboard consists of various image-associated buttons for different actions, designed to help illiterate farmers easily understand different functionalities. These interfaces are shown in Figure 6.2. The fourth dashboard, the Newsletter dashboard, is presented as the first figure in Figure 6.2.

6.2.3 Admin App

A separate admin app was developed which includes the signup and sign-in pages identical to those in the user app, along with the Admin Access interface featuring two text fields for Greenhouse ID and Email ID. A user authenticated by the admin for specifying their greenhouse ID along with their email ID can only access the Greenhouse Interface in the user app. The interfaces are as shown in Figure 6.3.

6.2.4 Crop Prediction

The crop prediction module have 3 different sub-modules ,with a manual recognition page in which all the inputs are entered by user,two different automation page based on climatic data,Soil Data in which first accepts climatic data via API and predicts the crop while the latter accepts climatic data via API and soil nutrient requirement shall be entered manually.The output shows the top 5 crops predicted for the place along with their nutrient requirement such as N,P,K required for 200lb and GDD .The 3 different pages are as shown in the Figure 6.4. The output predicted is shown in Figure 6.5.

6.2.5 Pest Identification

The Pest Identification module accepts image either by direct capture using camera or by uploading from gallery .The details of pest if present is predicted and is shown along with management details .The interfaces are shown in Figure 6.6.

6.2.6 Weed Identification

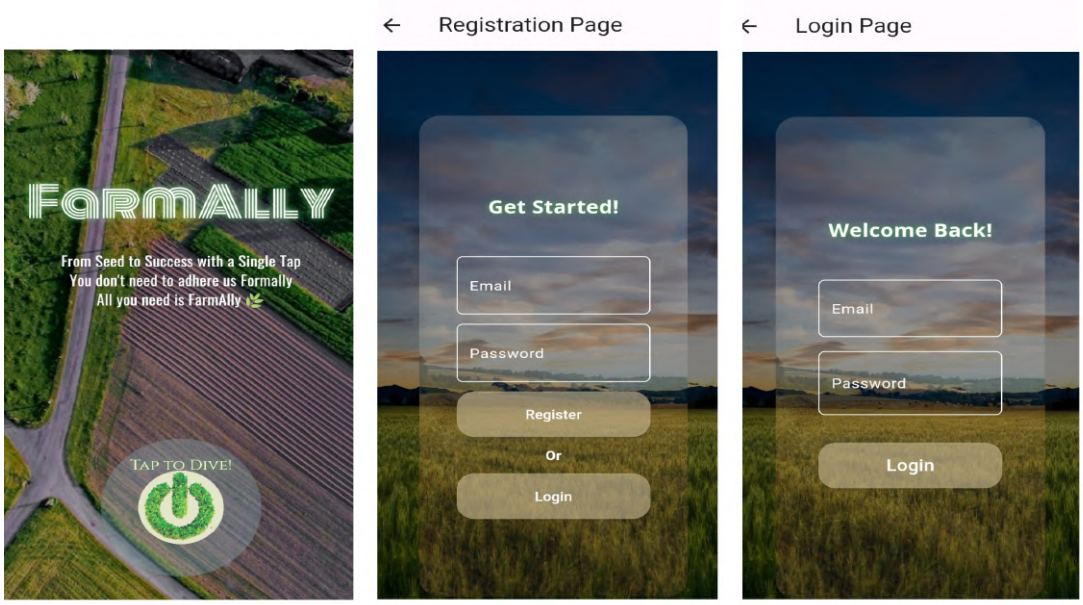
The Weed Identification module accepts image either by direct capture using camera or by uploading from gallery .The details of weed if present is predicted and is shown along with management details..The interfaces are shown in Figure 6.7.

6.2.7 Cost Estimation

The Cost Estimation module accepts 4 inputs from user the Forecast Year,State Name ,Crop Name and Area to be grown where State Name,Year and Crop Name provided as drop-down list .The predicted output is of The different costs required to grow the crop and the cost forecasted from 2024-2031 .The interfaces are shown in Figure 6.8.

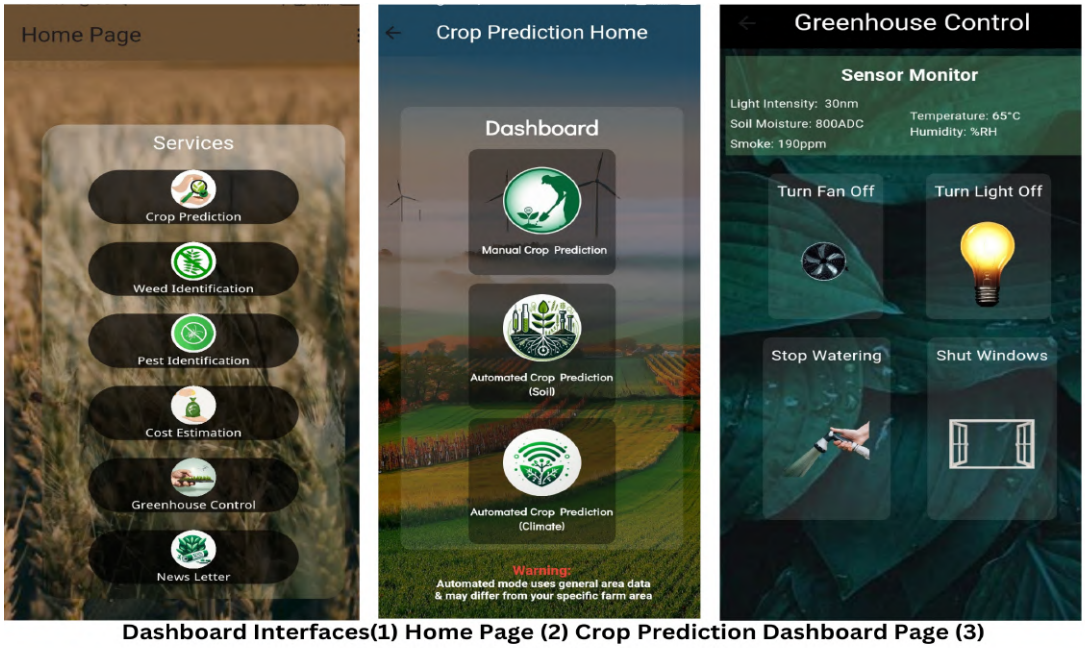
6.2.8 Newsletter

The Newsletter corner is a module added additionally as an advancement to the existing system to help farmers stay informed about current agricultural news and bookmark the news they find relevant. It features a News Corner section displaying all agricultural news related to India, fetched using the News API. Users have options for sharing or adding articles to favorites. If any news article is added to favorites, it will be displayed in the Favorites section. The interface is as shown in the Figure 6.9.



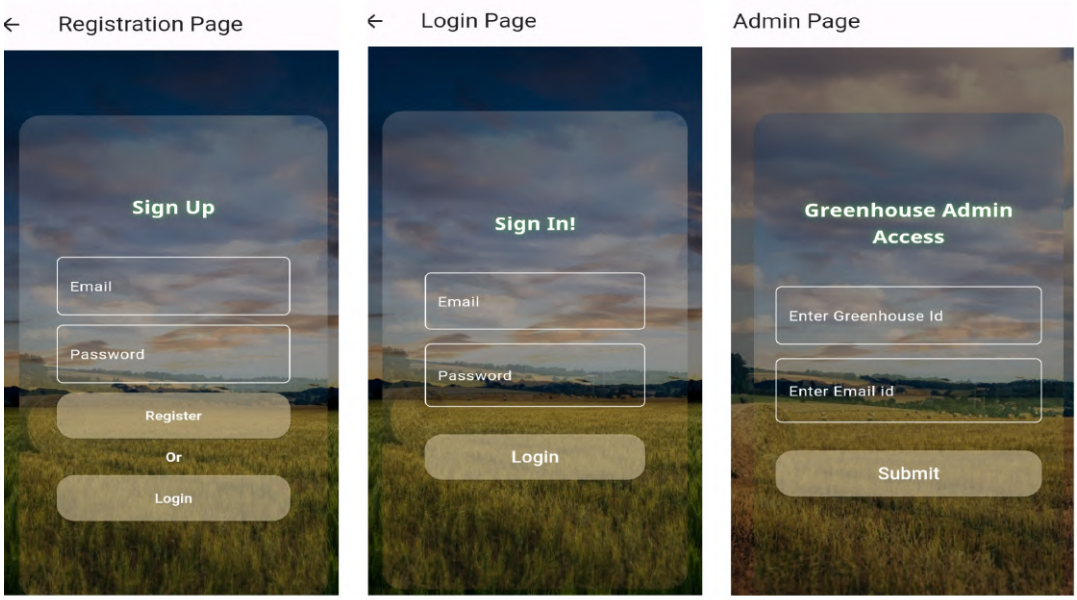
(1) App Page (2) Sign Up/Registration Page (3) Login Page

Figure 6.1: Authentication Interfaces



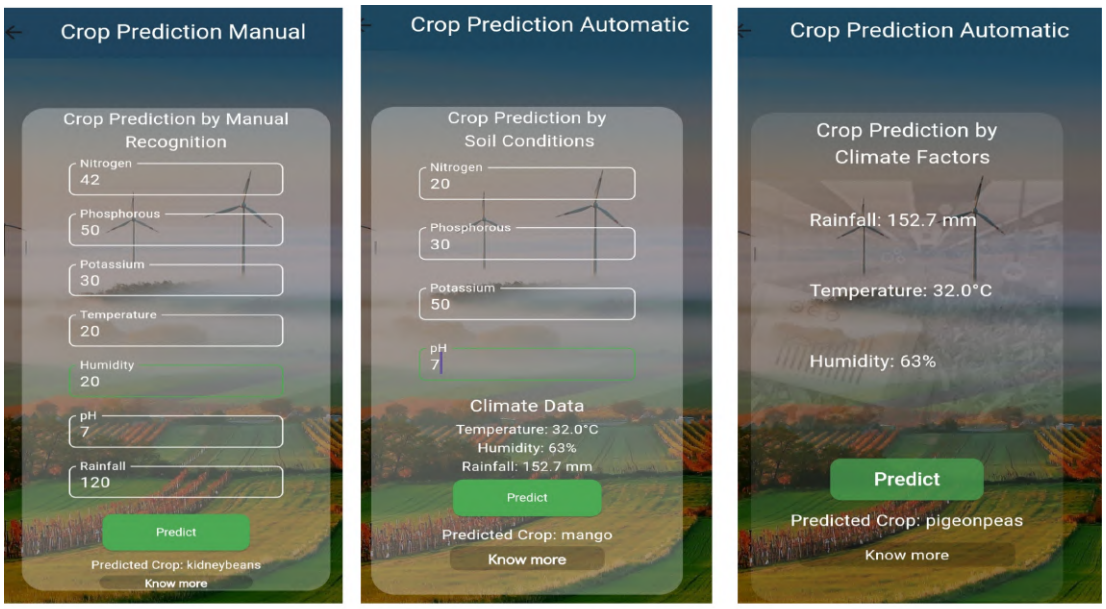
Dashboard Interfaces(1) Home Page (2) Crop Prediction Dashboard Page (3) Greenhouse Monitor Dashboard Page

Figure 6.2: Dashboards



Admin App Interfaces(1) Sign In/Registration Page (2) Sign Up/Login Page (3) Greenhouse Admin Access Page

Figure 6.3: Admin Page

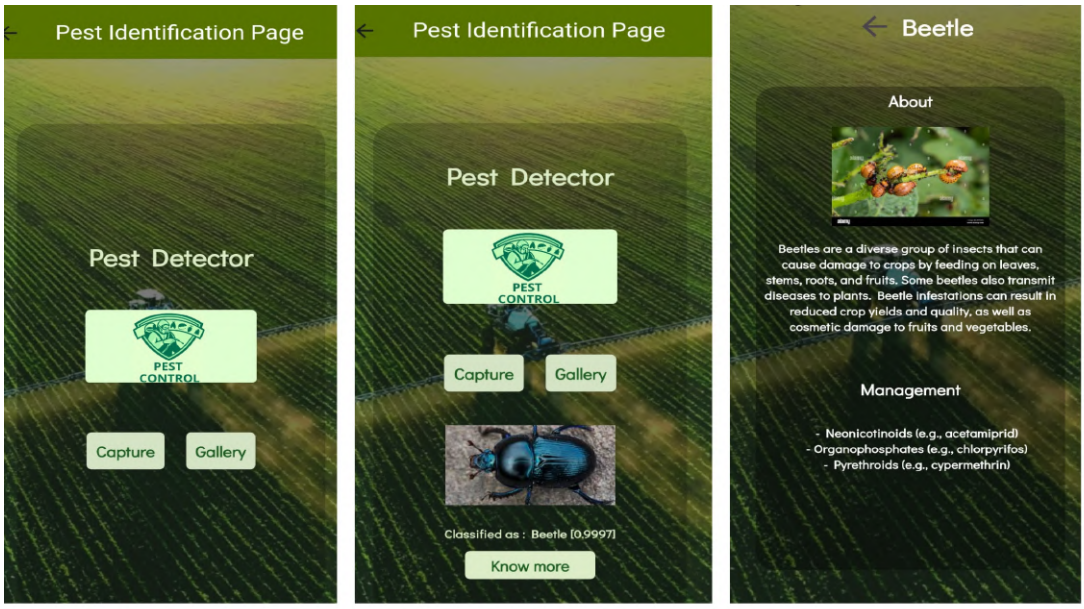


Crop Prediction page (1)Manual recognition (2) Partially automated climate and soil recognition(3)Completely automated recognition with only climate factors

Figure 6.4: Crop Prediction Page

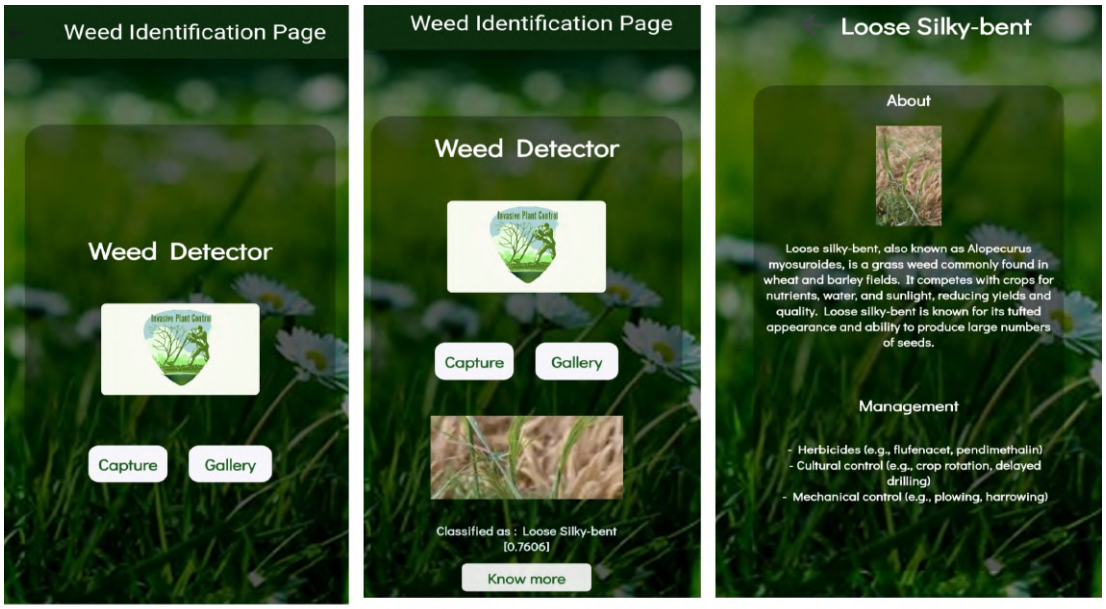


Figure 6.5: Crop Prediction Output



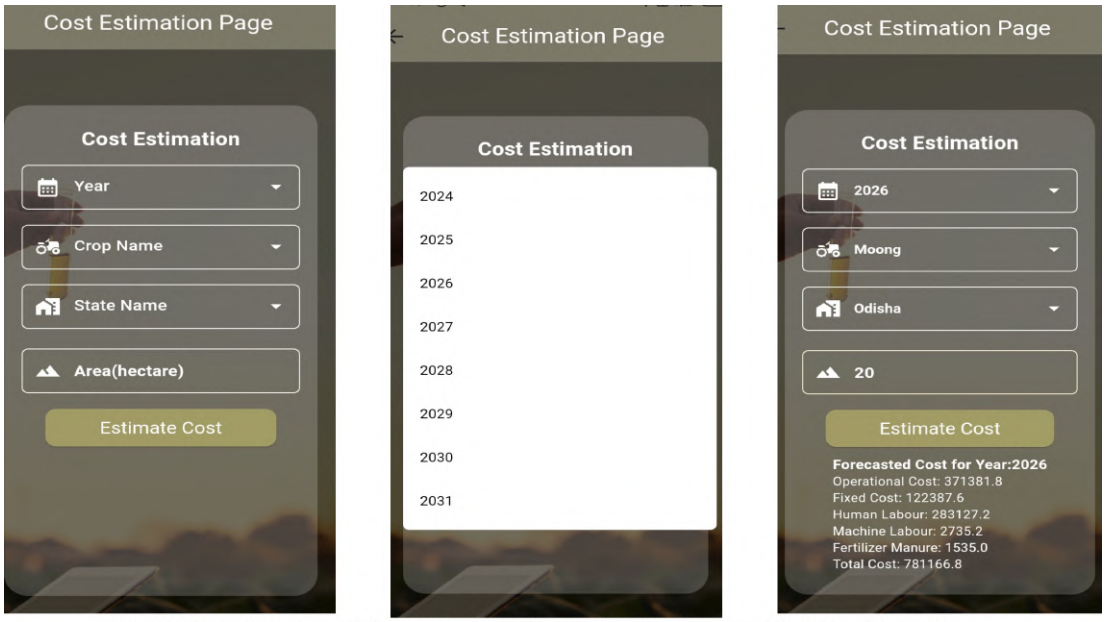
Pest identification page (1)normal interface (2) Pest Prediction result interface (3)Pest details and Management interface

Figure 6.6: Pest Prediction Page



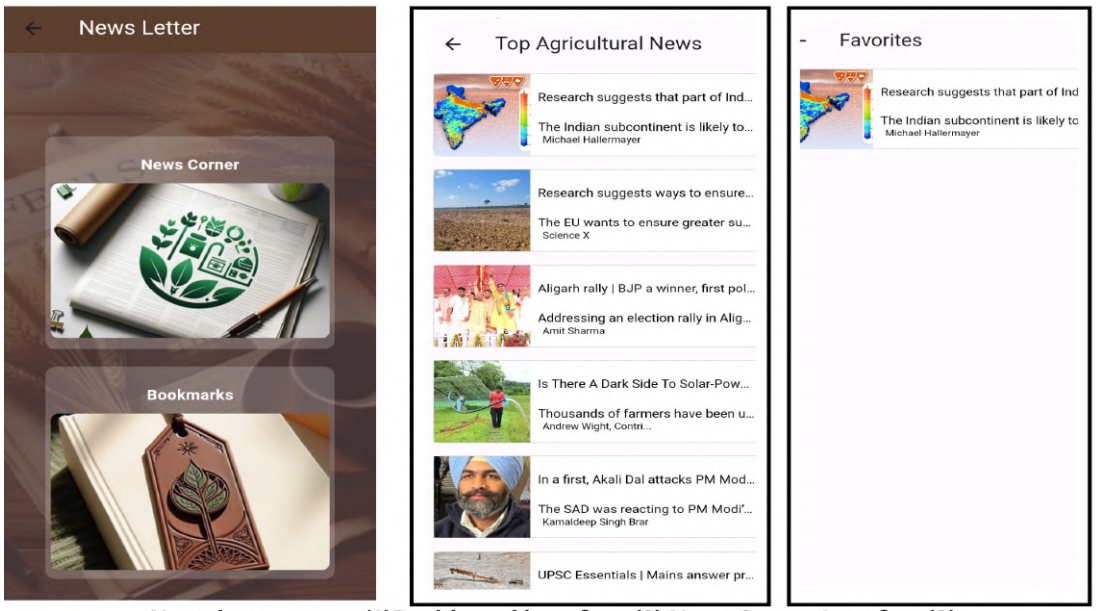
Weed identification page (1)normal interface (2) Weed Prediction result interface (3)Weed details and Management interface

Figure 6.7: Weed Prediction Page



Cost Estimation page (1)normal interface (2) Drop-down selection for input (3) Cost Forecast result interface

Figure 6.8: Cost Estimation Page



News lettter page (1)Dashboard interface (2) News Corner Interface(3) Favourites Section/Bookmarks interface

Figure 6.9: News Letter Page

6.2.9 Green House Control

The sensors use in the Greenhouse system are shown in the Figure 6.10.

Cooling Fan

The cooling fan works on the basis of temperature,humidity and smoke's presence threshold value.The Fan is turned on when the value of either temperature,humidity and smoke exceeds or above the threshold value and vice-versa as shown in Figure 6.11. with MAX6675 Temperature sensor dedicated for sensing temperature and MQ2 sensor dedicated for sensing smoke.

Sliding Window

The sliding window works on the basis of temperature,humidity and smoke's presence threshold value.The sliding window is opened when the value of either temperature,humidity and smoke exceeds or above the threshold value and vice-versa as shown Figure 6.12. with MAX6675 Temperature sensor dedicated for sensing temperature and MQ2 sensor dedicated for sensing smoke.

Light Control

The Grow Light works on basis of the basis of light intensity fetched from LDR sensor.The Light bulb is turned on when the value of light intensity is below the threshold value and vice-versa as shown in Figure 6.13.The module also stops working during the night by fetching IST time .

Water Pump

The water pump works on the basis of soil moisture's threshold value captured from Soil Moisture sensor.The Water pump is turned on when value soil moisture is inefficient and above threshold value(soil moisture sensor have opposite narrative sensing) and vice-versa as shown in Figure 6.14.

6.2.10 Notification System

The Notification System which gets activated on basis of different actuations in Greenhouse is as shown in Figure 6.15.The notifier only notifies when any of the modules(Fan,Grow Light,Water Pump,Sliding Window) are turned to on-state due to changes in values of different sensor threshold values associated with these modules while the user doesn't currently uses App.

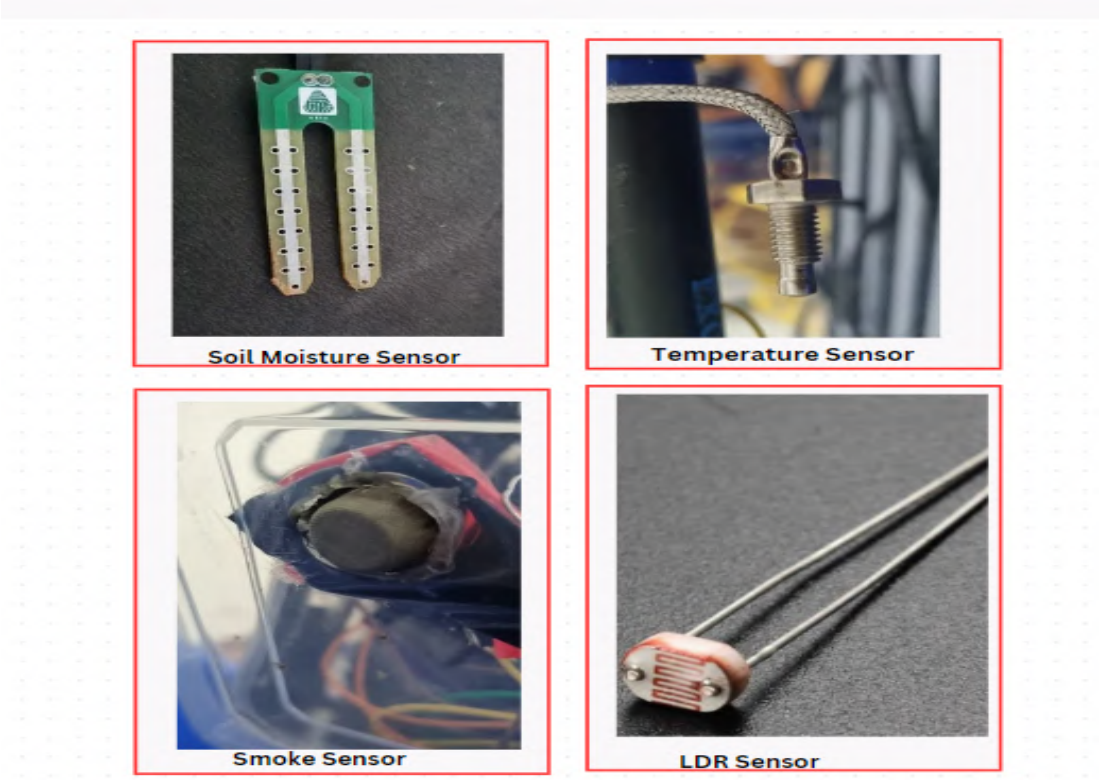


Figure 6.10: Sensors



Figure 6.11: Fan Automation

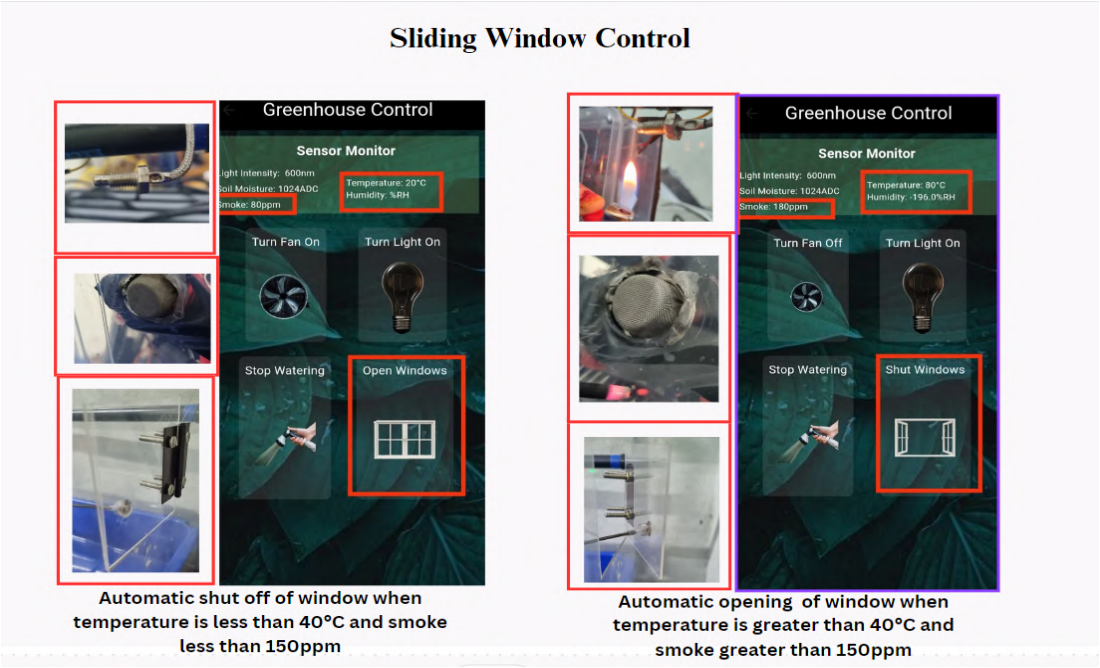


Figure 6.12: Window Automation

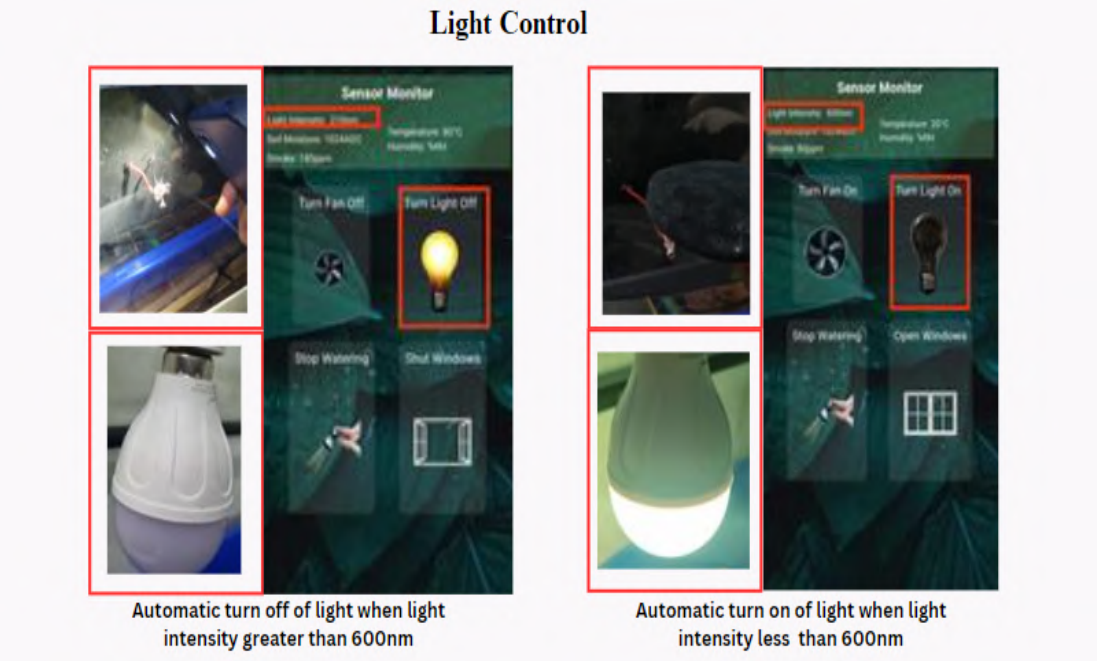


Figure 6.13: Light Automation

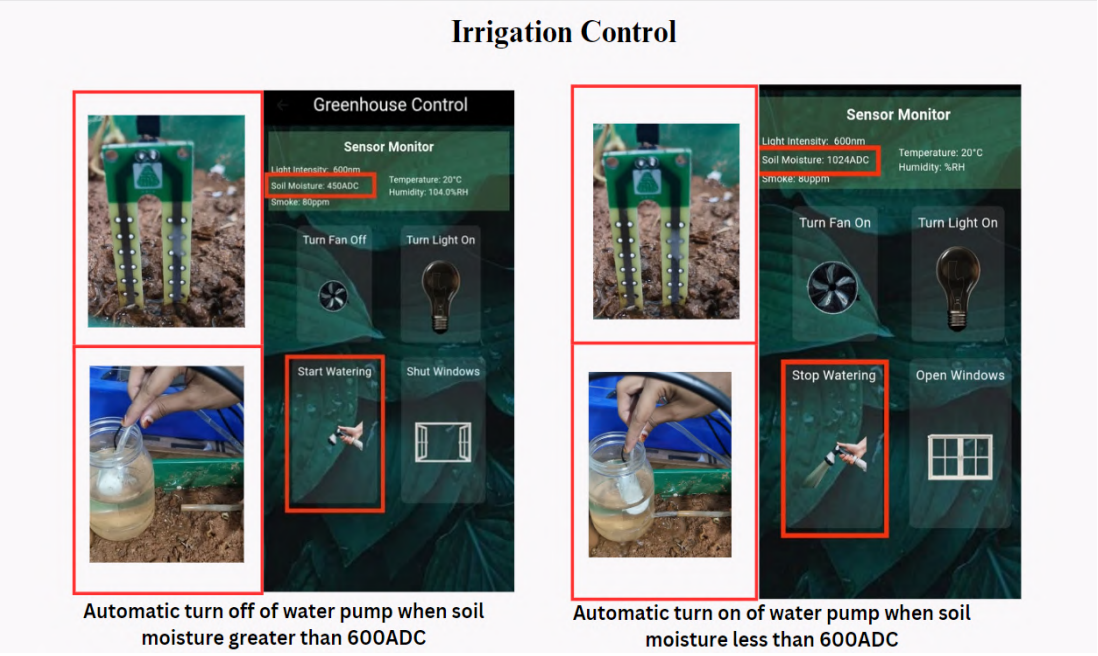


Figure 6.14: Irrigation Automation

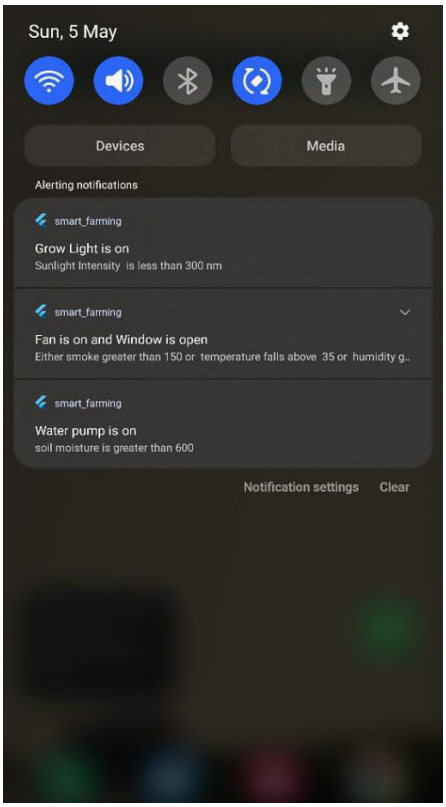


Figure 6.15: Notification System

CHAPTER 7

TESTING

7.1 Functionality Testing

Separate test cases are designed and tested for all the functional modules defines in the software requirement section.

- Verified Mobile Application Interface and all the pages corresponding to the system.
- Verified the Machine-learning prediction module of Crop Prediction with enhanced multiple choices for automation and manual recognition.
- Verified the Machine-learning prediction module of Cost Estimation
- Verified the Deep-learning prediction module of Weed Identification
- Verified the Deep-learning prediction module of Pest Identification
- Verified the manual and automatic working of modules of plant growth light system, Irrigation Control,Sliding Window control,Irrigation System.
- Verified the Notification system for the greenhouse working features.

7.2 UI Testing

The system allows only registered users to use the interface. The UI contains a login page, a registration page, an interface page where the users can divert to different features and a corresponding pages for each of the modules of Crop prediction,Cost Estimation,Pest Identification with additional pages for displaying detailed results as well as Cost Estimation and Green House Control page.

- Verified user authentication.
- Verified the interface page and the fetched detailed results page.
- Verified the smooth rendering of UI change corresponding to each of the actuations generated in Greenhouse
- verified the working of notification bar while the app is not running in background.

7.3 Integration Testing

The testing is done after all the modules of the system are completely developed, tested individually, and integrated. All the bugs during the development and testing are fixed and verified. Integration testing included the testing of the back end and interaction with the mobile app interface.

- Verified Firebase Integration with flutter.
- Verified Microcontroller integration with Firebase.
- Verified Flask server connectivity and smooth rendering to Flutter.
- Verified API fetching integration with Flutter.
- Verified the Admin App authentication integration with the user app Greenhouse interface.
- Verified Notification integration while app is not running.

CHAPTER 8

CONCLUSION

The Smart Farming project embodies a significant leap forward in revolutionizing traditional agricultural practices through the fusion of IoT, Machine Learning, and Deep Learning technologies. By leveraging these advanced technologies, the project's mobile application equips farmers with intelligent insights, thereby empowering them to enhance productivity and sustainability in their farming operations. The system's diverse capabilities, ranging from crop prediction to weed and pest detection, underscore its commitment to optimizing crop cultivation while minimizing resource usage. Although certain limitations, such as the absence of automated actions like herbicide spraying and harvesting, are acknowledged, the project remains adaptable and poised for future growth and enhancement.Looking ahead, the project envisions broad-scale adoption across farming communities, educational institutions, and governmental initiatives, thus fostering widespread impact and transformative change in the agricultural sector. By prioritizing the needs of non-technical users, particularly farmers, the application serves as an accessible and user-friendly interface, facilitating seamless interaction and integration of cutting-edge technologies into everyday farming practices.In an ever-evolving agricultural landscape, the Smart Farming project stands as a beacon of innovation, ushering in a future characterized by resilience, efficiency, and sustainability. Its steadfast commitment to efficient data management and sustainable agricultural practices positions it at the forefront of agricultural innovation, poised to catalyze positive change and shape the future of farming worldwide.

APPENDIX

8.1 DEFINITIONS

1. **pH**: pH stands for ”potential of Hydrogen.” It is a measure of the acidity or alkalinity of a substance,often used in agriculture to determine the acidity or basicity of soil, which can affect plant growth.

2. **Wi-Fi**: Wi-Fi is short for ”Wireless Fidelity.” In smart farming, Wi-Fi is used to enable wireless communication between devices and sensors, allowing for data transmission and remote monitoring.

3. **IoT**: IoT stands for the ”Internet of Things.” In smart farming, IoT refers to the network of interconnected devices, sensors, and equipment that collect and exchange data to optimize farm operations and decision-making.

4. **GDD**: Growing Degree Days are a measurement used by farmers and agricultural professionals to quantify heat accumulation and track the progress of plant growth and development. It’s a useful metric for determining the timing of various agricultural activities such as planting, harvesting, and pest management. GDD is calculated based on daily temperature data and provides insights into the thermal conditions that crops experience during their growth cycle.

5. **GPRS**: GPRS stands for ”General Packet Radio Service.” It’s a mobile data service that can be used for remote monitoring and data transmission in areas with cellular network coverage. Smart farming systems may utilize GPRS to transmit data from sensors and devices in the field.

6. **HTTP**: HTTP stands for ”Hypertext Transfer Protocol.” In smart farming, HTTP is often used to facilitate communication between devices, sensors, and servers. It’s essential for transmitting data over the internet and web-based control interfaces.

7. **GPS**:GPS stands for ”Global Positioning System” is a tool that helps you find the latitude and longitude of your current location or any address,In the project scenario GPS is used for fetching weather details of a place from an API.

8. **UI**:UI stands for ”User Interface”. It’s the graphical layout of a website or app that users interact with during navigation

9. **API**: API stands for ”Application Programming Interface.” In smart farming, APIs enable different software and hardware components to interact with each other. They allow third-party applications to access and utilize data and functionalities from the smart farming system.

10. **SDK**: SDK stands for ”Software Development Kit.” In smart farming, an SDK provides a set of tools, libraries, and resources to develop custom software applications that can integrate with or extend the functionality of the smart farming system.

11. **GDD**: Growing Degree Days are a measurement used by farmers and agricultural professionals to quantify heat accumulation and track the progress of plant growth and development. It’s a useful metric for determining the timing of various agricultural activities such as planting, harvesting, and pest management. GDD is calculated based on daily temperature data and provides insights into the thermal conditions that crops experience during their growth cycle.

12. **N** : It stands for element Nitrogen.The Nitrogen in the context specifies the Nitrogen present in the soil of land intended for agriculture.

13. **P**:It stands for element Phosphorus.The Phosphorus in the context specifies the Nitrogen present in the soil of land intended for agriculture.

14. **K**: This stands for element Pottasium.The Pottasium in the context specifies the Nitrogen present in the soil of land intended for agriculture.

15. **ML**:ML stands for ”Machine Learning” .It’s an Automating learning from data to make predictions without explicit programming.In the project’s context it’s used for Crop Prediction and Cost Estimation.

16. **DL**: DL stands for ”Deep Learning” is subset of Machine Learning (ML) specifically focuses on training neural networks with multiple layers to learn complex patterns from data.In the project’s context it’s used for Weed Identification and Pest Identification .

17. **NodeMCU**:NodeMCU stands for Node Microcontroller Unit.Open-source firmware for ESP8266, enabling IoT prototyping.This micro-controller is used as the sole source for programming all IoT components

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